CellFE Infinity MTx Operating Manual

Placeholder: Image of Infinity MTx

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Chapter 1. Product Information

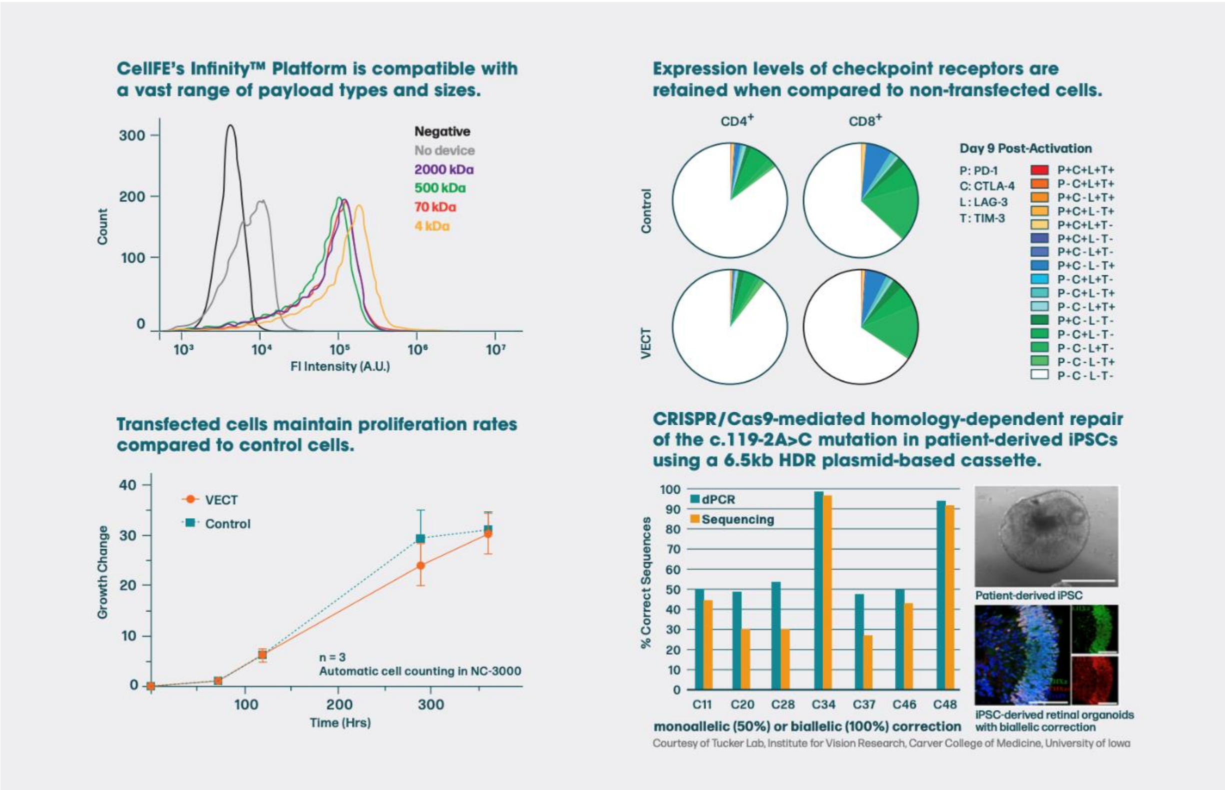
Product Description

CellFE's Infinity Platform is designed to deliver a wide range of payloads into any cell type, even with the most complex editing protocols needed to develop increasingly advanced cell therapies. This technology effectively delivers mRNA, CRISPR, Cas9 protein, and plasmid DNA and consists of the following two components:

> CellFE's Infinity MTx™ platform is a microfluidic transfection methodology that offers a novel way to transfect cells rapidly and simply. The platform successfully delivers a wide range of payloads into any cell type, even with the most complex editing protocols needed to develop increasingly advanced cell therapies. This technology has demonstrated effective delivery of mRNA, CRISPR Cas9 protein, and plasmid DNA while maintaining pre-transfection proliferation rates, cellular phenotypes, and cell subset ratios.
>
> The Infinity Microfluidic Processing Assembly (MPA) is a single use device wherein the transfection process occurs. The MPA contains sterilized membranes and an actuating lid that allows for aseptic use of the Infinity MTx platform. When the Infinity MTx platform directs the flow of cells though microfluidic channels, they pass under a single chevron ridge. This change in channel size leads to rapid cellular compression causing a fleeting loss of intracellular fluid and a subsequent decrease in cell size. Upon re-expansion, within milliseconds, the target payload is actively transported into the cell. This is known as VECT. The MPA is designed to process 8 separate channels of 25-100 µL and up to 2 million cells.

Key Benefits

- **Gentle on cells** — non-viral, non-electroporation technology
- **Exceptional performance** – high cell viability and delivery efficiencies with decreased recovery and expansion times
- **Simple, streamlined workflow** — easy to optimize and no need for cell concentration and buffer exchanges
- **Operational and cell-friendly processing** — no specific media, buffers, or chemical additives required



Product Contents

With the purchase of the Infinity MTx, CellFE provides the following products and accessories:

System contents

Product	Specifications	Quantity
Infinity MTx Instrumentation	CellFE 8-channel MPA compatible model	1
	SJT 16 AWG / 3C, Rating 125V 13A, NEMA 5-15P power cord (USA)	
	¼" NPT Pressure line tubing	
	2 x push to connect fitting	
Pressure source	*For in-house air:* Barbed hose to 1/4" NPT pressure line tubing and Norgen B07-2M1-M1KG filtered regulator	1
	No in-house air: P15-TC 115V Air Compressor, with push-to-connect fitting for 1/4" NPT pressure line tubing and Norgen B07-2M1-M1KG filtered regulator, and standard Teflon tape	
CellFE Infinity MPA (Optimizer kit)	2 x Infinity Optimizer kit A - 4 x MPAs with 2x chips of channel gap 5.2-5.8um	1 kit (4 MPAs)
	2 x Infinity Optimizer kit B - 4 x MPAs with 2x chips of channel gap 6.5-8.0um	
	2 x Infinity Optimizer kit C - 4 x MPAs with 2x chips of channel gap 9.0-12.0um	

Optimizer kit contents

Upon purchase of the Infinity MTx, users are asked to select an Optimizer Kit with sizes relevant to the cell types that will be used (see sizes below). Optimizer kits come with four MPAs. Once the desired MPA size is determined, users can purchase single-use MPAs of the desired size(s) in packs of four.

Optimizer Kits

Item	Optimizer kit A		Optimizer kit B		Optimizer kit C	
	Quantity #1 [item number]	Quantity #2 [item number]	Quantity #1 [item number]	Quantity #2 [item number]	Quantity #1 [item number]	Quantity #2 [item number]
Devices	1	5	1	5	1	5
Reactions	8	40	8	40	8	40

Item	MPA 5.2		MPA 5.5		MPA 5.8	
	Quantity #1 [item number]	Quantity #2 [item number]	Quantity #1 [item number]	Quantity #2 [item number]	Quantity #1 [item number]	Quantity #2 [item number]
Devices	1	5	1	5	1	5
Reactions	8	40	8	40	8	40

Item	MPA 6.0		MPA 6.2		MPA 6.7	
	Quantity #1 [item number]	Quantity #2 [item number]	Quantity #1 [item number]	Quantity #2 [item number]	Quantity #1 [item number]	Quantity #2 [item number]
Devices	1	5	1	5	1	5
Reactions	8	40	8	40	8	40

Item	MPA 7.2		MPA 7.7		MPA 8	
	Quantity #1 [item number]	Quantity #2 [item number]	Quantity #1 [item number]	Quantity #2 [item number]	Quantity #1 [item number]	Quantity #2 [item number]
Devices	1	5	1	5	1	5
Reactions	8	40	8	40	8	40

Item	MPA 9		MPA 10		MPA 11	
	Quantity #1 [item number]	Quantity #2 [item number]	Quantity #1 [item number]	Quantity #2 [item number]	Quantity #1 [item number]	Quantity #2 [item number]
Devices	1	5	1	5	1	5
Reactions	8	40	8	40	8	40

System Components

Infinity MTx device

Front Panel

1. **User Interface Touchscreen –** From here, the Operator can detail and conduct the transfection processes, view and export information about previous transfection runs, change default system settings, and access additional information about operating the Infinity MTx. (See the software section of the manual for more information).

2. **Sample Loading Tray** – Compatible with the Infinity MPA and Optimizer kit. Use the Eject Button on the touchscreen to open the tray. The MPA or Optimizer kit can be placed in the open tray by matching up the protrusion on the device with the recess on the tray.

   WARNING: Ensure nothing is touching the seam between the tray and the main body of the instrument, as this can damage the instrumentation and may cause harm to the user.

3. **Barcode Scanner –** Located on the underside of the instrument above the tray. The barcode scanner reads the barcodes on the Infinity MPA package and saves device specific information to the notes section. Press the "Scan /Read ID" Button to scan the barcode.

Placeholder: Image of opened tray (either from angle or birds-eye view)

Placeholder: Image of someone scanning the barcode on the device packaging

Back Panel

1. **Exhaust Air Vents** – Used to vent air out of the instrument. **CAUTION** Do not block – Ensure at least 6” of clearance behind instrument
2. **Compressed Air Inlet** – The primary attachment point of the instrument –compatible with ¼” NPT Pressure line tubing fitted with a push-to-connect fitting. The instrument should be attached to a pressurized air source of 100 PSI for optimal functioning. The pressurized air should not exceed 120 PSI.
    a. Connecting the tubing: Ensure the air source is turned off and decompressed. Gently push the push-to-connect fitting into the compressed air inlet of the instrument.
    b. Disconnecting the tubing: **WARNING**: Severe injury can occur if the instrument air line is

Placeholder: Labeled image of Back Panel of the Infinity MTx, with numbers corresponding to below sections

disconnected prior to turning off the instrument, turning off the supply of instrument air and depressurizing the line. Turn off the instrument and shut off the air source prior to performing this step. Ensure both systems are decompressed before removing the tubing by pushing the small tab located on the instrument's compressed air inlet.

Placeholder: Labeled image of the Infinity MPA with numbers corresponding to below sections

Placeholder: image of power switch

3. **Air Intake Fan** – Provides cooling for internal electrical components and power supply. **CAUTION** Do not block – Ensure at least 6" of clearance behind instrument
4. **Power Connection Port** – Auxiliary port that powers the instrument. The C13 to 5-15 power cord can be plugged directly into the Infinity MTx, with the opposing 3-prong end able to connect to a properly grounded 3-prong, 110-120V/50-60Hz power source.
5. **Power Switch** – Switch that turns the instrument on and off. Electrical supply is limited by the Power Switch, and after operation it is best to shut the instrument off.
6. **USB-A Connection Port** – An auxiliary port that a USB type A can be attached to. Used to export data. This data export process is detailed in the software section.

   LAN Connection Port – An auxiliary port that a LAN cable can be connected to. For current customer use, the instrument does not have internet connection capabilities.

WARNING: Do not attempt to connect the Infinity MTx to the LAN. CellFE will not be held responsible for any damage that results as the Infinity MTx is not meant to be connected to your network.

Infinity MPA

1. **MPA Lid** – Attached via a hinge to the collection chamber lid. Once samples are loaded into the sample chambers, the lid can be closed by pushing down until the lid latch is locked in place. Once closed, the membrane filters located on the air inlet and outlet ports allow filtered compressed air to be introduced aseptically.
    i. **MPA Lid Latch** – A structural feature that allows for reversible locking of the MPA lid to the collection chamber lid. To open the lid, gently apply pressure to the bottom of the MPA lid latch while holding the sides of the collection chamber lid. After samples are loaded, the lid latch can be closed by gently applying pressure to the top.

    ii. **Air Inlet and Outlet Ports** – These are the Infinity MPA's primary interaction points with the Infinity MTx instrument. These ports allow for the aseptic in-flow and out-flow of pressurized air to each of the 8 sample chambers. These ports each contain a filter membrane.

2. **Collection Chamber Components**

    Making up the lower portion of the Infinity MPA, the collection chamber houses eight reservoirs that capture the output from the transfection process. The outer portion of the collection chamber contains latches that allow for the attachment and detachment of the collection chamber lid. Users can either detach the entire top portion of the MPA to collect samples, or just open the MPA lid.

    i. **Collection chamber lid**– Attached to the collection chamber and MPA lid. The collection chamber lid covers the eight sample chambers, which house the microfluidic chip microchannels. The collection chamber lid and MPA lid can be detached from the collection chamber by gently squeezing the collection chamber latches.

    ii. **Sample chamber** – Each sample chamber can be loaded with up to 100 uL of sample volume. The 8 sample channels are also designed to fit the dimensions of a standard multi-channel pipette to allow the user to quickly fill all channels of the Infinity MPA. **Note:** For maximum yield, ensure no droplets appear on the edges of the sample chamber when loading the samples.

    iii. **Microfluidic Chip Microchannels (2 channels per chip) –** The microfluidic chip is where the transfection process occurs. Each sample chamber contains one microfluidic chip, and each chip contains two channels. Within each channel is a ridge like structure called a "chevron". Between the channel and the ridge is a gap, and this gap size determines the type of cell that will be transfected through the given channel. **CAUTION** Sample chambers should not be reused for subsequent transfection processes.

    iv. **Collection Chamber Latches –** Latches that enable the separation of the collection chamber and the collection chamber lid. Removing the lid can allow for easier access to and removal of the transfected cell samples. The collection chamber lid can be removed by holding the lid and pushing in both latches while gently pulling the collection chamber lid off.

Chapter 2. Upon Receiving the Infinity MTx

Site Preparations

Physical space requirements

- The Infinity MTx shall be placed on a level laboratory bench or within a tissue culture hood.
- The Infinity MTx requires a space of at least 16.75” x 16.75” x 15.75” for operational clearance.
- The area surrounding the instrument must be clear and the ventilation slots must be unblocked to allow for adequate air flow.
- Sufficient room should be left behind the instrument to allow access to the Power Switch and rear connections. Ensure that there is enough space to prevent connection lines and cables bending more than 45°.
- Select a location with relative humidity and temperature as specified in Appendix E for optimal instrument performance.

Electrical Requirements

- For the power connection, select a location close to a 110-240 VAC, Frequency 50/60 Hz outlet.

Infrastructure Requirements

- The Infinity MTx requires a compressed air source of at least 100 PSI for operation.
- If facility has a compressed air connection available, ensure that the physical space and electrical requirements of the instrument can be met by the air source. Contact CellFE for assistance in setting up the line from the available air source if needed.
- If requested, CellFE can provide a Werther P15-TC 115V Ultra-quiet oil-lubricated air compressor. The compressor has the following specifications:
    - Space requirements for compressor: 12” x 14” x 15” for operational clearance.
    - Electrical requirements for compressor: One 110-240 VAC, Frequency 50/60 Hz receptacle.
    - Ensure compressor and attached lines are placed in a location such that they won’t cause a tripping hazard.
    - Refer to the Werther P15-TC 115V User Manual for detailed information.

Safety and environmental considerations

- The Infinity MTx is designed for indoor use only. Temperature, relative humidity, and altitude recommendations are provided in Appendix E.
- During installation, personal protective equipment (PPE), including safety goggles or glasses, disposable gloves, and a lab coat (if working in a biosafety level 2 facility or higher), must be utilized.
- Do not cover the ventilation openings. Provide at least 20 cm of clearance to allow proper airflow and to allow access to the Power switch.

- Wait to turn on the primary air source/air compressor until it is directly attached to the Infinity instrument.
- Only remove the line connecting the Infinity MTx and air source when air source is turned off, the Infinity instrumentation is off, and the pressurized components in the instrumentation are adequately relieved.
- **WARNING**: Severe injury can occur if the instrument air line is disconnected prior to turning off the instrument, turning off the supply of instrument air and depressurizing the line.

Setting Up the Infinity MTx

Unpacking and Inspecting

1. Inspect the package for damage prior to opening. Do not unpack if damaged. If damaged, take photos and submit them to the shipping company and to CellFE for disposition.
2. If the package is undamaged, open it. Remove the Infinity carrying case, ensuring it is placed face up on the ground next to the installation site. Confirm that the site provides sufficient clearance for the instrument, is level, and free of debris.
3. Carefully remove the first two layers of foam from the carrying case.
4. Lift the instrument out of the case using both hands and place it on the site, ensuring the screen is facing the correct way up and all four rubber feet of the instrument contact the placement surface.
5. Inspect the outer wrap of the Infinity MTx for any gaps, movement, and/or shifting of the panels. Ensure that the instrument screen is fixed and secure, with no gaps between the screen and the casing. Verify that the tray is closed and aligned with the instrument wrapping.
    5.1. **Note:** If any issues are observed, refer to the troubleshooting section of this document before proceeding.
6. If applicable, unpack the air compressor following these steps:
    6.1. Open the box containing the air compressor and remove the surrounding packing foam, oil, and accessories from the compressor box, setting them aside.
    6.2. With both hands, lift the compressor by the handle, remove it from the box, and place it on the ground in a safe location.
    6.3. Replace the rubber stopper on the outlet tube with the provided air filter.
    6.4. Prepare the push-to-connect fitting by wrapping it with 2 layers of Teflon tape as described below:
        6.4.1. Leave the first 2 threads bare, then wrap the fitting clock-wise two times.
    6.5. Then gently screw it into the Norgen pressure regulator.
    6.6. Wrap the screw end of the Norgen pressure regulator with 2 layers of Teflon tape (as described above) and use an adjustable wrench to tighten the pressure regulator to the outlet of the air compressor.
    6.7. Fill the compressor with oil following these steps:
        6.7.1. Connect one end of the small diameter tubing to one end of the larger diameter tubing provided with the compressor.
        6.7.2. Remove the remaining rubber stopper from the compressor.
        6.7.3. Insert the other end of the small tubing into the open port.

Placeholder: Image of rubber stopper being replaced with filter

6.7.4. Remove the screw cap on the bottle of oil, carefully remove the film, and replace the screw cap with the provided screw cap with a nozzle. Remove the red tip, and using scissors cut the tip of the nozzle. Place the tip into the other end of the large tubing.
6.7.5. Gently squeeze the bottle to fill the compressor with oil until sight glass is half filled.
6.7.6. Once the appropriate oil level is reached, remove the oil bottle, cap the nozzle with the red tip, place the tubing in a small Ziplock bag, and close the port with the rubber stopper.

Placeholder: Image of someone removing the rubber stopper and placing tubing assembly into port.	Placeholder: Image of someone filling the compressor tank with oil.	Placeholder: Image of bauble empty vs. half full.

Assembling and Connecting the MTx

1. Once the Infinity MTx is in place, attach the tubing to the compressed air inlet, located on the back panel of the Infinity MTx, ensuring a secure connection. **Note:** If the fitting is not going into the compressed air inlet, trip the lock by pushing in the tab located on the compressed air inlet.
2. Attach the other side of the tubing to the primary air source or air compressor.
3. Turn on the primary air source and if necessary open any valves along the primary air source line.
4. Lift and turn the top of the Norgen pressure regulator and regulate the pressure according to the dial to 100 PSI.
5. Plug in the power cable into the power connection port located on the back panel. Then connect the other end of the power cable to a 110-240 VAC, Frequency 50/60 Hz receptacle.

Testing MTx Functionality

1. Use the Power Switch (which is on the back panel) to power up the Infinity MTx.
2. While the system is booting, add 30 uL of deionized or molecular water to each of the 8 sample chambers in the liquid transfer test device.
3. Once the instrument is ready, select the "Demo" protocol on the main screen.
4. Open the sample loading tray by using the Eject Button, load the liquid transfer test device into the Infinity, and withdraw the tray by pressing the same button. Press "Next" to proceed.
5. Select the "Installation" protocol and press "Next" to proceed.
6. On the "Run" screen, press the "Start" Button to initiate the liquid transfer process.

7. Observe the Run Messages dialogue box (for normal operation, only the "Run Start" and "Run Complete" messages should appear). If there are any run errors, refer to the Infinity status and Run Errors section otherwise, remove the test device from the Infinity MTx after operation is complete.
8. Open the liquid transfer test device to see if the liquid transfer process completed successfully.
    8.1. **Note:** If the liquid has been transferred successfully (there are ~30 uL of water in each of the 8 collection reservoirs)

Placeholder: (optional) MPA loaded in all 8 sample chambers with 30 uL of dyed water

Placeholder: (optional) same MPA after processing showing water in sample collection reservoir.

Chapter 3: Using the Infinity MTx

Software user interface

Software comes pre-installed on the Infinity MTx. See the images and descriptions below for a full understanding of each window and touchscreen icon.

Upon turning on the instrument, the title screen will appear (image below). This is also the screen that the idle instrument defaults to.

Figure 1- Title Screen

Touch anywhere on the screen to proceed to the Operator Selection Screen.

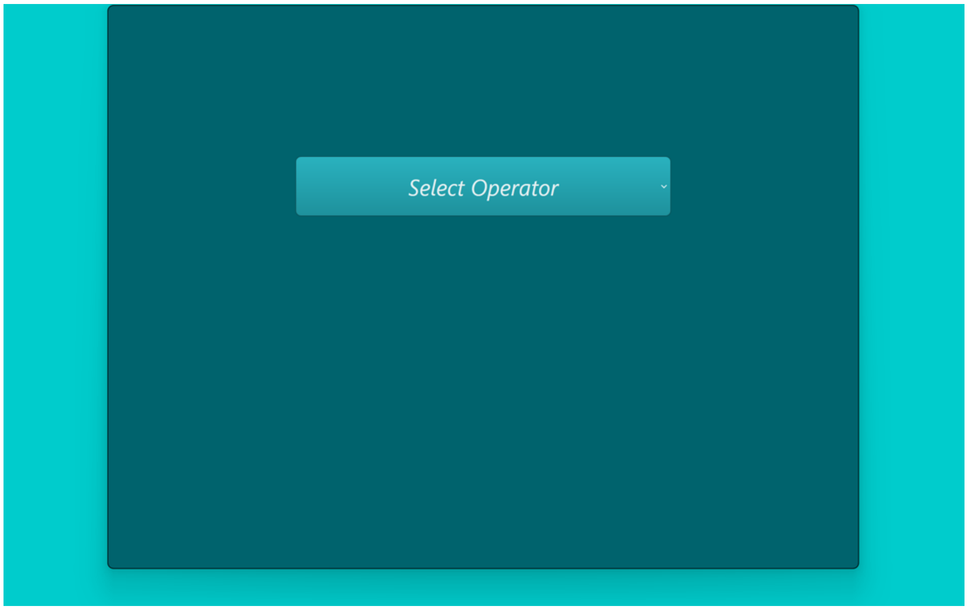
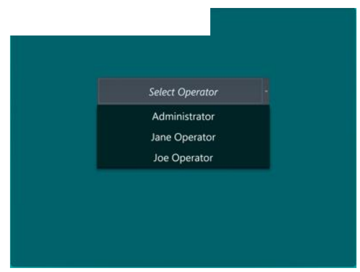


Figure 2 - Operator Selection Screen



The Operator Selection Screen is used to select the user of the Infinity MTx. To initiate the selection, open the menu by pressing "Select Operator".

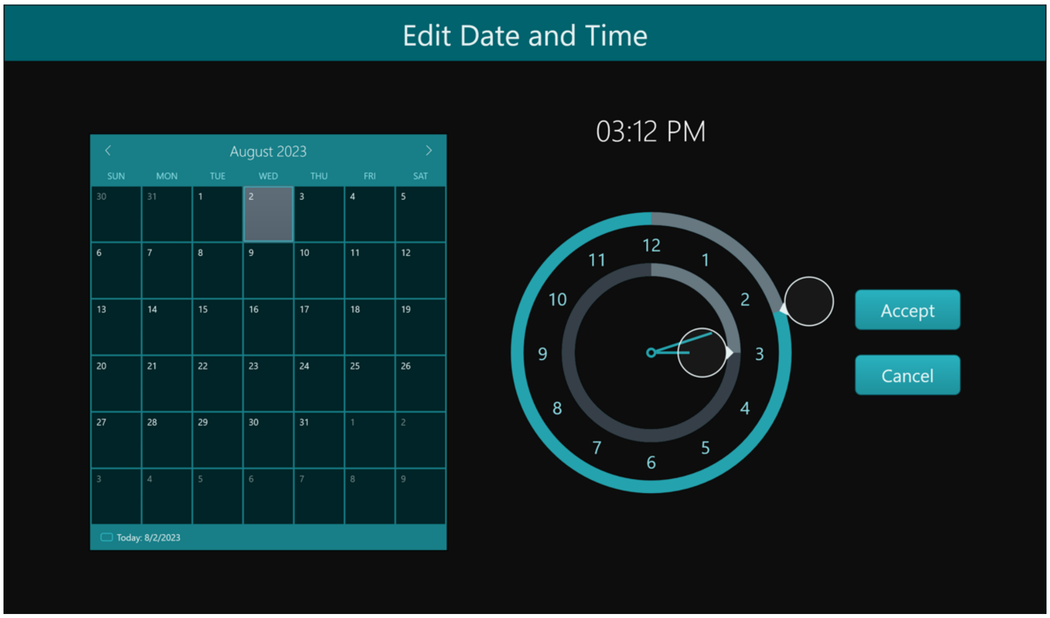
Managing users

There are two kinds of users: Operators and Administrators (Admins). Admins have access to "administrator settings", which allow the Admin to perform a variety of system wide changes (including editing protocol information, changing date and time, and adding, editing and deleting users).

"Edit Operators" Button – This button allows the Admin to add, edit and delete Operators. **NOTE:** This will not remove the name from previously logged runs. Press the "OK" Button to return to the Administrator Settings Menu.

"Edit Laboratory Name" Button – This button allows the Admin to create a system name for the Infinity MTx. This name will be linked to the Infinity and will be present through the protocol setup screens.

"Edit Date and Time" Button – This button allows the Admin to view and update the system's date and time. To select a date, navigate to the month by using the arrow keys on the calendar and then select a specific date on the calendar. To select a time, press the bubbles on the clock, the inner circle dictates the hour, and the outer circle dictates the minute. After the date and time is set as desired press" Accept" to return to the Administrator Settings Menu. **Note:** Date/time values are linked to each run in the run history screen.



After selecting the operator, users will be prompted to enter their password.

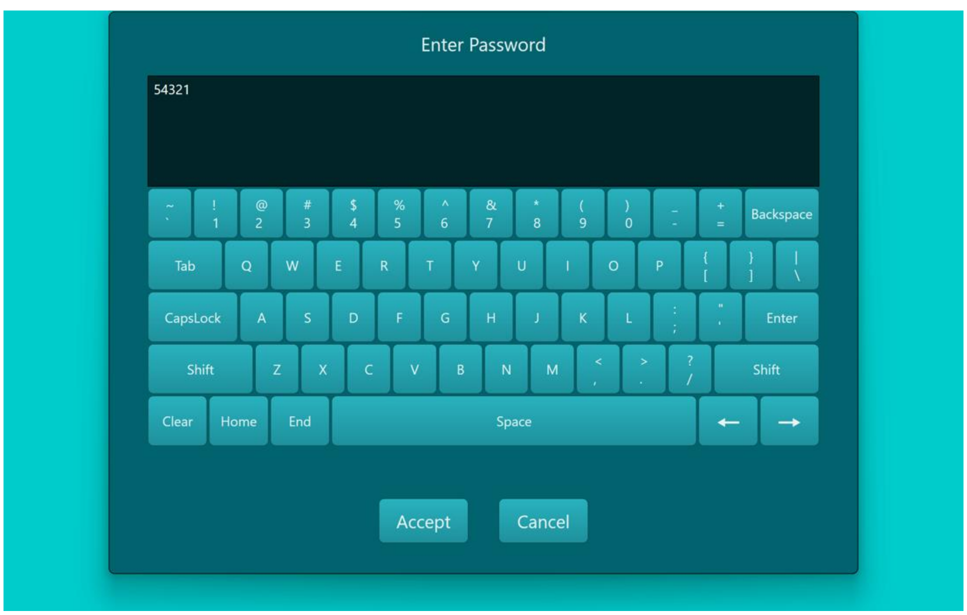


Figure 3-Virtual Keyboard

The virtual keyboard allows the user to easily input text.

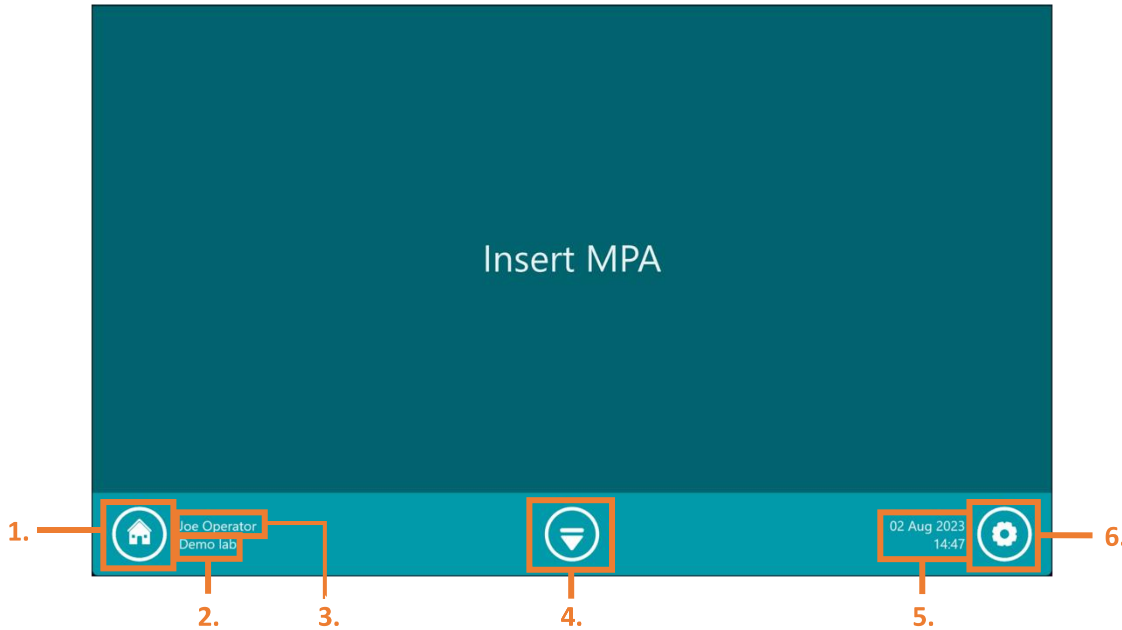
Touchscreen Controls

Icon	Function
	Go to Home Screen.
 Withdraw In movement Eject	Withdraw allows user to withdraw tray, in movement shows tray in use, and eject allows tray to be ejected.
	Go to System Settings.
	

Setting Up and Running a Protocol

Three steps are required for the set-up of the protocol: 1) loading the MPA, 2) selecting the desired protocol, 3) changing aspects of the protocol to suit the given experiment.

1) Load the MPA



Icons and Tools

Icons and tools found on the Protocol Setup Screen are as follows:

1. **Home Button –** This button (right) is present throughout a variety of instrument operations. Pressing it will return the Operator to the Operator Selection Screen, where if the user desires to continue any operations will be required to login again using their specific password.

2. **Current Operator Name –** Displays the name of the current Operator as selected from the Operator Selection Screen.

3. **Tray Eject/Withdraw Button** – This button is present during all operations and allows the Operator to eject and/or withdraw the tray to ultimatly insert the Infinity MPA device. The button has three states, withdraw, in movement, and eject, depending on the current position of the drawer (right).

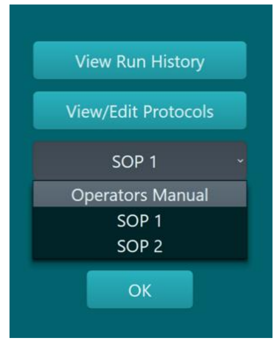
4. **Local Time and Date** – Displays the local time and date as a running clock. This time/date is aligned with the timestamps of each transfection process run (see run history section). The time and date can be changed by the System Administrator in the Administrator Settings.

5. **Settings Button –** This button is present throughout a variety of instrument operations and allows the Operator to access the system settings. This includes options to “Edit Protocols”, “View Run History”, access Administrator Settings [“Administrator Settings”] (only shows if the administrator has selected their name

on the Operator Name Selection Menu and input their password), view information “About” the instrument. Use “OK” or “Exit” to return to screen before Settings Button was pressed.

a. **“View Run History” Button -** This button allows the user to access the list of previous run operations performed on the Infinity MTx. The Operator can see the date and time of each run, the protocol it was performed with, the particular device used, and any run notes or messages. The Operator is also able to identify descriptive information from each run including which sample chambers were ran, and which cell types or payload types were used. Additionally, the Operator can use previous runs as a template for new protocols. See Run History section for more details.

b. **“Edit Protocols” Button -** This button allows the user to access the list of saved protocols on the Infinity MTx and be able to edit existing protocols, add new protocols, or delete protocols (Administrator only). See Edit Protocols Section for more details.

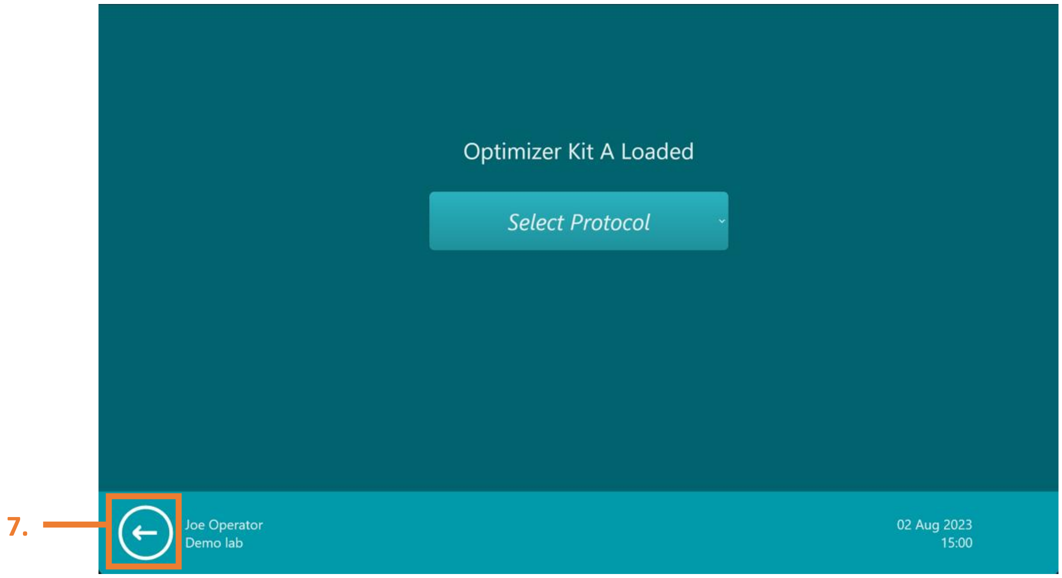


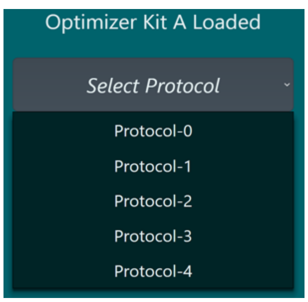
c. **“Documents” Button –** This button allows the user to access a list of documents related to the Infinity MTx operation (right). This includes, but is not limited to, documents such as this operation manual, and specific protocols for various cell types and applications.

d. **“Administrator” Settings Button –** If the user is logged in as the Administrator, administrator settings can be accessed by pushing this button. This provides the Administrator with a new menu with a variety of functions, including the options to “Edit Payloads”, “Edit Cell Types”, “Edit Activation Times”, “Edit Operator Names”, “Edit Laboratory Names”, and “Edit Date and Time”. Editing these functions enables the Administrator to be able to change some of the pre-set options or characteristics for other users during the normal Infinity MTx operations. Press “OK” to revert to the settings menu.

e. **“About” Button –** This button allows the user to access the Infinity MTx software and firmware version details (right). Press “OK” to revert to the settings menu.

2) Select the desired protocol





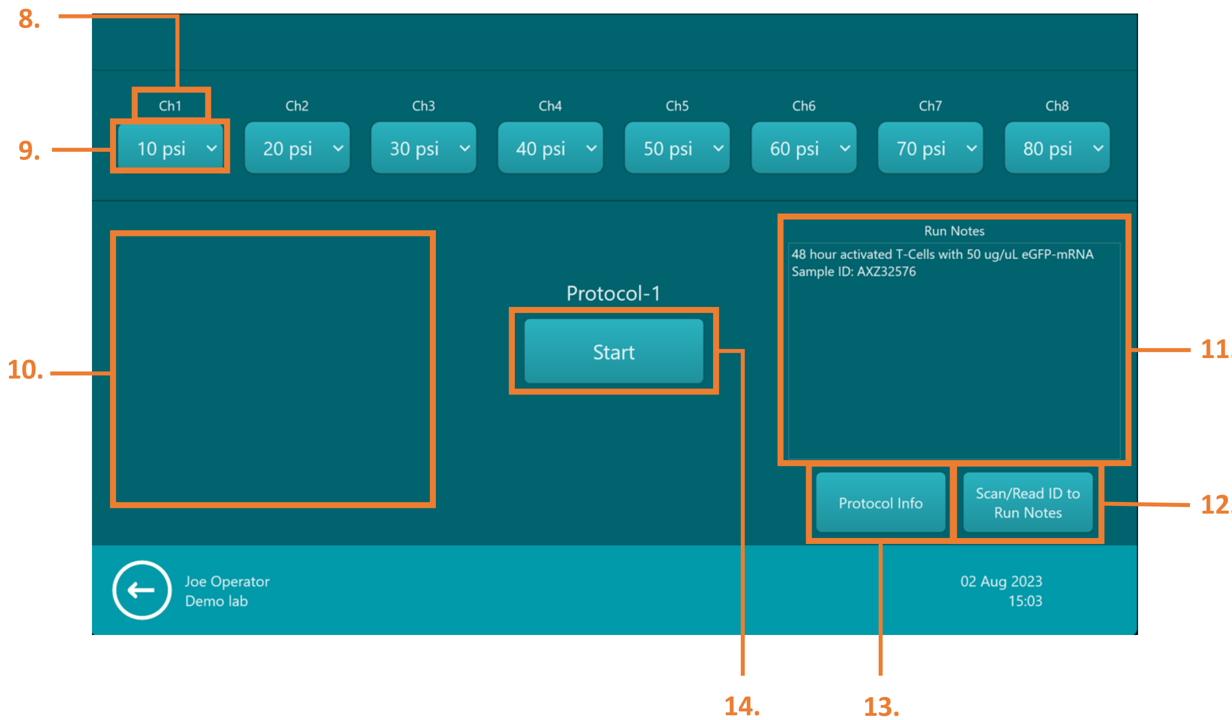
Upon loading the MPA into the Infinity MTx the instrument will read a variety of information associated with the MPA, including consumable type, lot number, date of manufacture, shelf life, the max applicable pressures that can be used with the channels, the channel status (i.e., which channels have been used before), the last date the channel was used, and the use life. Based on some of this information the Infinity MTx will prompt the user to select a protocol, specific to that device type, by pressing the drop-down menu located in the middle of the screen (left).

6. **Back Button –** Pressing the back button will return the user to the previous screen. (i.e., in the case of being on the Protocol Selection Screen, if the Back Button is pressed the user will return to the insert MPA Screen.)

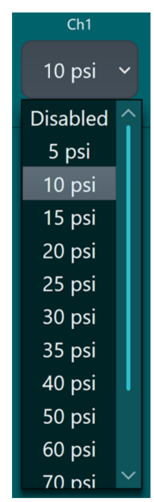
Once a specific protocol has been selected by the user, the Infinity MTx will proceed to the Run Screen, where specific parameters of the experiment can be selected, and the transfection run itself can be initiated and monitored.

3) Choose run settings

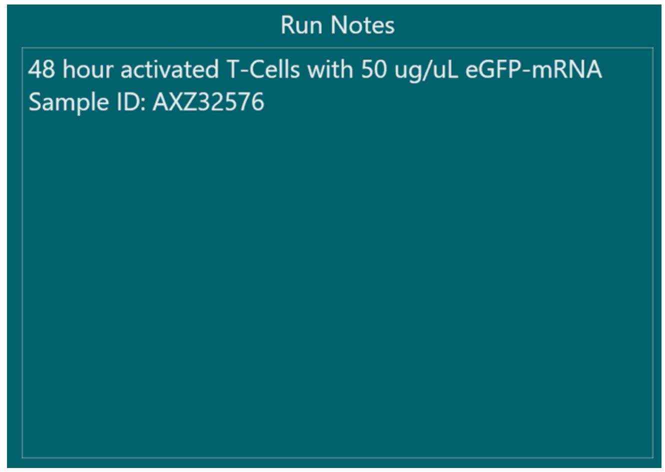
Once the Infinity MPA is loaded into the MTx, the Operator will be presented with the Run Settings Page. This page contains components that dictate the transfection processing parameters, dialogue boxes to indicate run notes or messages, and sections to input information. The Run Settings Page is also used to initiate the transfection process and track the process' status.



8. **Sample Chamber Number –** Displays the distinct sample chamber number. This channel is assigned to a corresponding chamber number on the Infinity MPA.
9. **Sample Chamber Pressure Select Menu –** This button allows the Operator to select the supply pressure in pounds per square inch [PSI] for the transfection process for each of the 8 distinct sample chambers. After pressing this button, the Operator will be presented with a scrollable menu containing a list of pressure values that can be used for the transfection process (right). This menu allows the Operator to select pressure in increments of 5 or 10 between within the range of 5 PSI and 100 PSI. Additionally, the pressure can be set to "Disabled", which will prevent the sample chamber from being processed during a run.



10. **Run Messages Dialogue Box –** This dialogue box displays run messages that can occur during operation (below, left). This includes information about errors that can occur with specific channels both before and during the transfection process has initiated. Messages will appear as they occur with the timestamp indicated for each message.

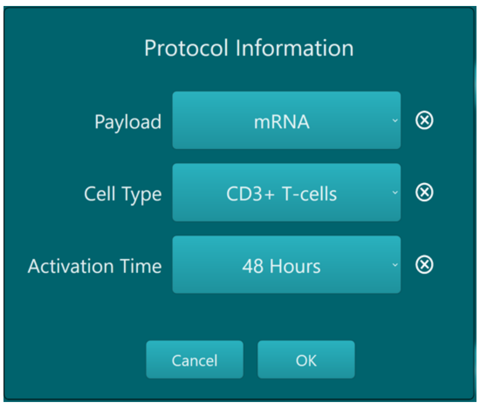


11. **Run Notes Dialogue Box –** This dialogue box displays run notes that can be written by the Operator before the transfection process has initiated. **NOTE:** Notes can be written in this section after the transfection process has occurred, however, these notes will not be recorded in the Run History Log. Notes will be displayed as text in the box and are also viewable in the run history after the run has finished (above, right). Press the Run Notes Dialogue Box to pull up the Infinity MTx's virtual keyboard. After filling out the notes press "Accept" to return to the Run Settings Page. All notes will be saved when the run is started.

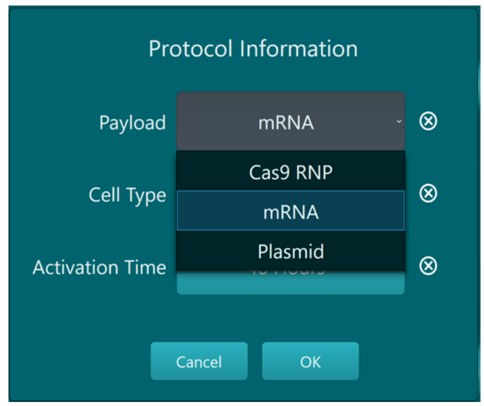
12. **"Scan /Read ID" Button –** Pressing this button will initiate the barcode scanner functionality and during this time barcodes available on the Infinity MPA packaging can be scanned. Scanning the barcode will input the specific device information into the Run Notes Section. Additional notes can be added to the Run Notes Dialogue Box .

13. "Protocol Information" Button – Pressing the protocol information button allows the Operator to input various information, including the payload, cell type, and activation time, specific to their desired run. While this information has no effect on the run parameters, this optional section to fill in additional information will be included and viewable in the Run History Section attached to the associated run. Pressing the button while pull up a pop-up box with three drop-down menus, "Payload", "Cell Type", and "Activation Time" (below). Each of these menus will have a variety of default values. **NOTE:** Only Administrators can add, edit, and delete values from this list.

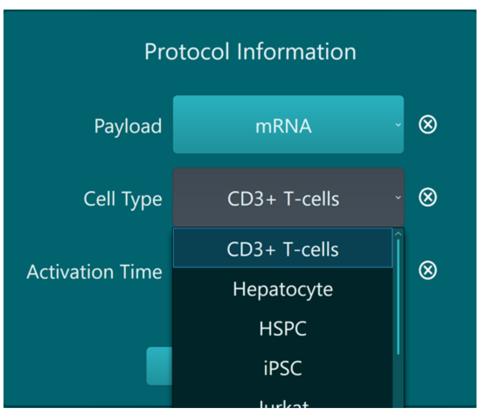
Additional Information Menu View



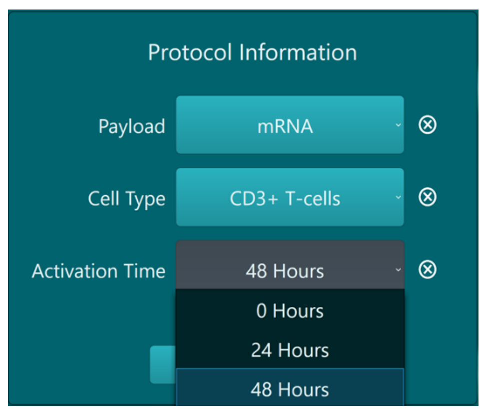
Additional Information (Payload)



Additional Information (Cell Type)



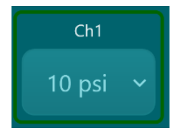
Additional Information (Activation Time)



Running the protocol

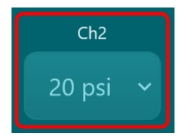
14. "Start" Button – Pressing this button will start the transfection process by initiating the Infinity MTx's engagement with the Infinity MPA device and supplying pressure to each of the 8 sample chambers where pressure supply has been enabled. **Note:** After pressing Start, the channels and pressures cannot be changed; ensure processing parameters are set and run notes and additional information are provided where desired before pressing Start. Sample chambers will be processed from 1 to 8 in numerical order. When beginning the transfection process, colored borders will appear around each channel to signify the status (below).

a. **Green Border -** Status indicator shows that the specified channel is on and the sample in the sample chamber has completed the transfection process with no errors.



b. **Yellow Border -** Status indicator shows that the specified channel is on, pressure is being applied to the channel, and the sample in the sample chamber is currently undergoing the transfection process.

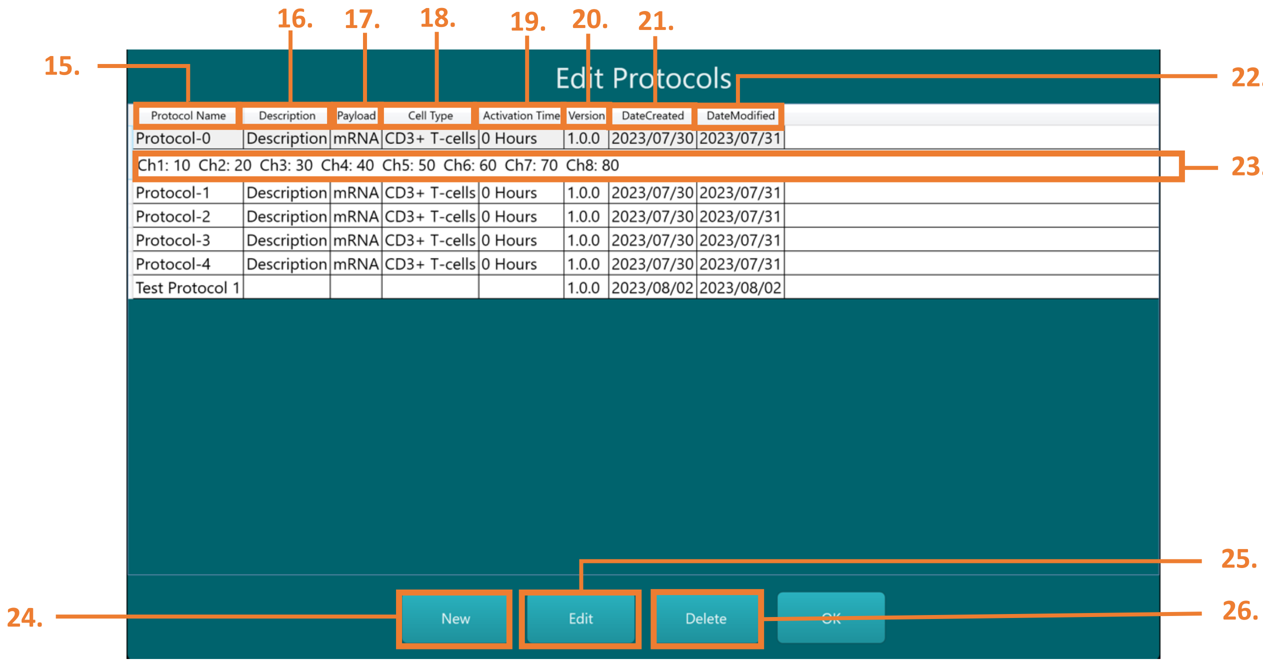
c. **Red Border -** Status indicator shows that the channel encountered an error while processing or the run was manually aborted before the liquid transfer process was complete. **NOTE:** error messages will appear in run messages dialogue box.



After all samples are processed, the "Remove MPA" Button will be displayed (right). The Operator can press the Tray Eject Button to open the sample loading tray and remove the MPA. The Infinity will then prompt the use to "Insert the tray to continue". Upon pressing the Withdraw Tray Button the Infinity MTx will save the run data (includes run notes, run messages, and the transfection parameters) to the Run History and return to the Insert MPA Screen.

Protocol List View

The Edit Protocols Screen is accessible via the Settings Menu for both Operators and Administrators. This protocol list view allows the user to quickly view the parameters of protocols saved on the Infinity MTx. In addition to viewing protocol details, the user can create new, edit existing, and delete protocols. Return to the Insert MPA screen by pressing "OK".



15. **"Protocol Name" Column –** This column shows the name of each specific protocol as made by the user during the initial protocol creation.

16. **"Description" Column –** This column shows the description of each specific protocol. During protocol creation, the user can use the virtual keyboard to input a brief description of the protocol.

17. **"Payload" Column –** This column shows the payload related to each specific protocol. During protocol creation or editing, the user can input a payload from a list of payload types (as set by the Administrator) using the "Protocol Information" Button.

18. **"Cell Type" Column –** This column shows the cell type related to each specific protocol. During protocol creation or editing, the user can input a cell type from a list of cell types (as set by the Administrator) using the "Protocol Information" Button.

19. **"Activation Time" Column –** This column shows the activation time related to each specific protocol. During protocol creation or editing, the user can input an activation time from a list of times (as set by the Administrator) using the "Protocol Information" Button.

20. **“Version” Column –** This column shows the version of the Infinity MTx that the protocol was created on.

21. **“DateCreated” Column –** This column shows the date that each specific protocol was created in the format “YYYY/MM/DD”.

22. **“DateModified” Column –** This column shows the date that each specific protocol was last edited by a user in the format “YYYY/MM/DD”.

23. **Detailed Protocol View –** By pressing on each protocol, the expanded and detailed view of each protocol will be shown. This expanded view details which channels are set to run with each specific protocol and what pressure [PSI] each channel is set to operate with. Close this view by pressing again on the protocol’s primary row.

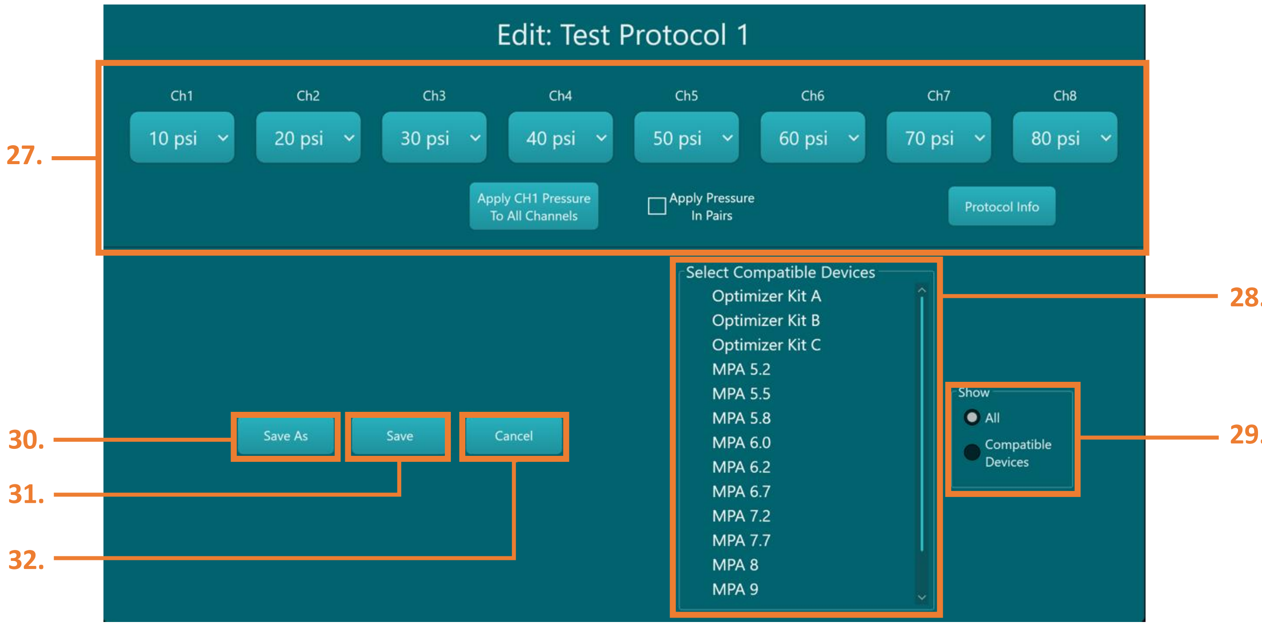
24. **“New” Button –** By pressing this button the user can create a new protocol. The user will be prompted to use the virtual keyboard to name the new protocol. After pressing “Accept”, the user will be taken to the Edit Protocol – Edit Screen (see Edit Protocols – Edit Screen section).

25. **“Edit” Button –** The user can press this button after selecting a pre-existing protocol on the Protocol List View. The user will be taken to the Edit Protocols – Edit Screen (see Edit Protocol – Edit Screen section), where they will be able to edit the selected protocol and save any changes.

26. **“Delete” Button –** The user can press this button after selecting a pre-existing protocol on the Protocol List View. A pop-up window will appear confirming if the user wants to delete a specific protocol. Once deleted the protocol will not appear on the Protocol List View.

Editing Protocols

When creating a new protocol or editing a pre-existing protocol, the user will be presented with the Edit Protocols – Edit Screen. This screen has many similarities to the Edit Run Settings page seen during protocol operation, but the primary purpose of this screen is to configure settings and information for specific protocols.



27. **Protocol Parameters and Details Selection –** This section provides pressure selection for each of the 8 sample chambers, like the Protocol Run Page. Additionally, this selection includes access to the "Protocol Info" Menu where payload, cell type, and activation time details can be selected. Additional functionality is also available including:

    a. **"Apply CH1 Pressure to All Channels" Button –** This button allows the Operator to quickly apply the pressure set for chamber 1 to all the chambers. Using this button will overwrite the current values on chambers 1 through 8.

    b. **"Apply Pressure in Pairs" Checkbox –** Pressing this checkbox will grey out chambers 2, 4, 6, and 8 preventing the use of the sample chamber select menu for these channels. If this box is checked, the Operator can input different or the same pressures in chambers 1, 3, 5, and 7. The values will be linked to the greyed-out chambers where chamber 2 will display the same pressure as chamber 1, 4 will display the same pressure as chamber 3, 6 will display the same pressure as chamber 5, and 8 will display the same pressure as chamber 7.

28. **"Select Compatible Devices" Menu –** Using this menu, the user can select one or multiple types of devices that the protocol can be compatible with. This compatibility refers to which protocols will be available in the Protocol Selection Screen after the device type has been inserted in the sample loading tray and the MPA information has been read by the Infinity MTx. For example, if the user sets up a protocol "X" to be compatible for only MPA 5.2 devices, protocol "X" will only show up when a MPA 5.2 device has been loaded and similar will not show up on the Protocol Selection Screen if another device type is loaded.

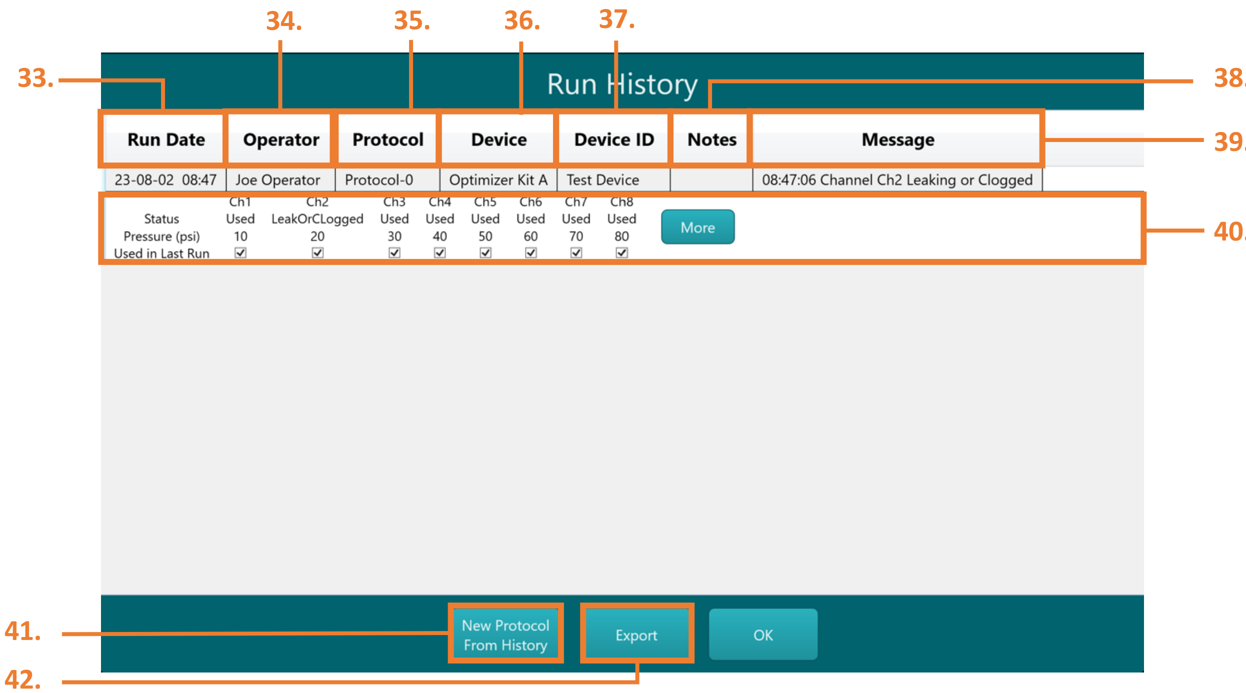
29. **"Show" Selection Box –** This selection box will limit the Select Compatible Devices Menu to display all the MPA devices available from CellFE or just the compatible devices as set by the user.

30. **"Save As" Button –** Pressing this button will prompt the virtual keyboard to show up. The user can then input a new protocol name and after pressing the "Accept" Button, the user will be taken back to the Protocol List View. This allows for the user to keep the original and unaltered protocol template and provides a new and edited protocol.

31. **"Save" Button –** Pressing this button will overwrite the original protocol template with the edited changes and will result in a name change. The user will be taken back to the Protocol List View.

32. **"Cancel" Button –** Pressing this button will cancel all existing edits to the protocol and return the user to the Protocol List View.

Viewing run history



Using the Settings Menu, the user can access the Run History Screen. This screen provides a variety of functions to the user, including the ability to access and view detailed information of previous runs, export information and data associated with previous runs, and the ability to create a new protocol from a previous run. This section will go over the details that the Run History Screen provides as well as the functions that are accessible on this screen.

33. **"Run Date" Column –** This column shows the date and exact time of each recorded run. The format is as follows "YY-MM-DD HH:MM".

34. **"Operator" Column –** This column shows the name of the Operator or Administrator that performed the run.

35. **“Protocol” Column –** This column shows the name of the protocol that was selected to perform the run.

36. **“Device” Column –** This column shows the type of device that was used during the previous run. This column refers to MPA categorization such as, “Optimizer Kit A” or “MPA 5.2”.

37. **“Device ID” Column –** This column shows the serial number associated with the device used for the previous run.

38. **“Notes” Column –** This column shows any notes written by the user during the previous run. **NOTE:** Only notes recorded prior to starting the run will appear on the Run History Screen.

39. **“Messages” Column –** This column shows any messages that occurred during the previous run. This will primarily include error messages that may have appeared during operation.

40. **Detailed Run View –** By selecting a run, the expanded and detailed view of the run will be shown. This expanded view details a variety of information including, which chambers were used in the run, as denoted by the checkbox, the run status associated with each channel (“Used”, “Unused”, or “LeakOrClogged”), and if marked “used” or an error occurred will show the specific pressure in PSI used with each chamber. More button? Close this view by pressing again on the associated row.

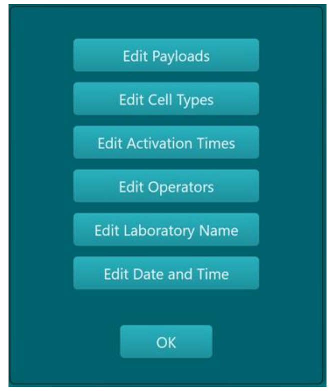
41. **“New Protocol From History” Button -** The user can press this button after selecting a previous run on the Run History Screen. The user will be taken to the Edit Protocols – Edit Screen (see Edit Protocol – Edit Screen section), where they will be able to edit the protocol used during the chosen run. Once edited as desired, the user can save any changes.

42. **“Export” Button –** If a user has selected one or multiple runs and a USB-A device is plugged into the USB-A connection port, located on the back panel of the Infinity MTx, the user can press the “Export” Button to transfer the data files and information to the USB drive, in the form of CSV files.

WARNING: It is recommended that the user scan any USB drive (i.e., to remove viruses and malware) prior to plugging it into the USB-A connection port.

Administrator Settings - Protocols

The Administrator can make a variety of system-wide changes, including editing the payloads, editing the cell types, and editing the activation times as they appear when the protocol info is filled out during a run.



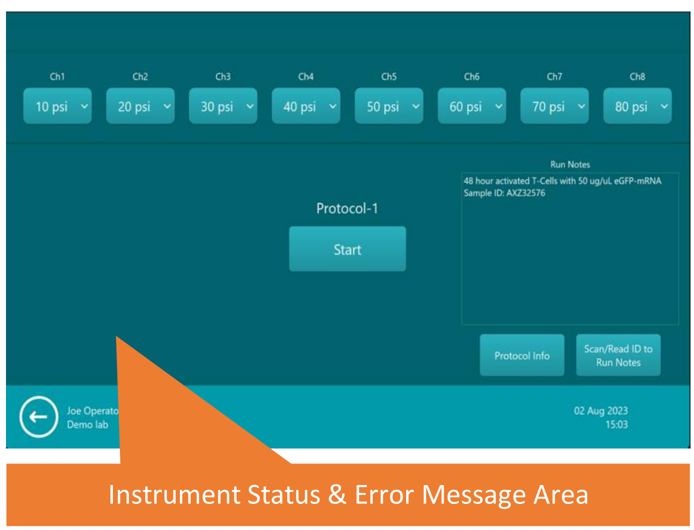
"Edit Payloads" Button – The edit payloads window will show all the available payloads that can be selected in using the "Protocol Info" button. Pressing the "Add New" Button will open a virtual keyboard where the user can type in the name of any new payload to be added. Pressing the "Delete" button after selecting a particular payload on the window will delete the payload on the system. Pressing the "Restore Default List" will return the list to the factory list of payloads. Press the "OK" Button to return to the Administrator Settings Menu.

"Edit Cell Types" Button – The Edit Cell Types Window will open and will show all the available cell types that can be selected in using the "Protocol Info" Button. Pressing the "Add New" Button will open a virtual keyboard where the user can type in the name of any new cell type to be added. Pressing the "Delete" Button after selecting a particular cell type on the window will delete the cell type on the system. Pressing the "Restore Default List" will return the list to the factory list of cell types. Press the "OK" button to return to the Administrator Settings Menu.

"Edit Activation Times" Button – The Edit Activation Times Window will open and will show all the available activation times that can be selected in using the "Protocol Info" Button. Pressing the "Add New" Button will open a virtual keyboard where the user can type in the name of any new activation times to be added. Pressing the "Delete" Button after selecting a particular activation time on the window will delete the activation. Press the "OK" button to return to the Administrator Settings Menu.

Status and Run Errors

Errors can occur throughout operation, and may appear as pop-up boxes in the Run Message Dialogue Box during sample processing. In order to mitigate the error, one must determine the cause and perform a corrective action. During sample processing, some errors may halt the liquid transfer process to prevent any instrument damage or damage to the MPA. Should an error arise, the status indicator for the relevant chamber will turn red and a message will appear in the Run Messages Area indicating the chamber where the error occurred and likely what the error was. In the case of significant error, the Infinity MTx will halt operation.



The table below can be used to identify particular issues and potential solutions. **NOTE:** Some errors that occur during sample processing may result in the loss of the sample.

Error Message	Possible cause	Action
Profile Errors		
Password incorrect. Please re-enter password or select different operator.	The user has entered the incorrect password for a selected operator profile.	The user should enter valid password or contact the Administrator to re-set user's password.
Protocol Setup – Insert MPA Screen Errors		
Presence sensor cannot detect consumable. Please insert device or return to Home.	The presence sensor is unable to recognize the device. The run cannot be continued until the presence sensor is able to recognize the device.	The user can re-open the tray and re-insert the MPA if this step was missed during workflow. If the presence sensor is broken, the user should return to the Operator Selection Screen and contact customer support.
NFC tag cannot be read. Please remove the MPA and scan the MPA barcode or return to Home.	The NFC tag reader is non-operational.	To mitigate the error scan the device barcode using the barcode reader located on the front panel. Otherwise, the user should return to the Operator Selection Screen and contact customer support.
NFC tag not valid.	The NFC tag is being read but is unable to be validated by system.	The user should return to the Operator Selection Screen and contact customer support to replace faulty device.
The barcode cannot be read due to issues with barcode scanner. Please return to Home.	The barcode reader is non-operational.	The user should return to the Operator Selection Screen and contact customer support.
Consumable not recognized. Please return to Home.	The device barcode is being read but is unable to be recognized by system.	The user should return to the Operator Selection Screen and contact customer support to replace faulty device.

Error Message	Possible cause	Action
Consumable has passed expiration date. Would you like to continue?	The device has passed the established expiration date.	User can choose to continue with the transfection process or to return to the Operator Selection Screen and utilize a new device. **NOTE:** Normal operation may be affected, and transfection efficiency and cell survival may be impacted
Consumable has passed "use by" date. Would you like to continue?	The device has passed biological expiration date (The amount of time after the first use of device has exceeded the recommended time).	User can choose to continue with the transfection process or to return to the Operator Selection Screen and utilize a new device. **NOTE:** Normal operation may be affected, and transfection efficiency and cell survival may be impacted
All channels have been used, consumable cannot be run.	All 8 of the sample chambers in the device have been previously used.	User must return to the Operator Selection Screen and utilize a new device.
Protocol Setup – Editing Run Settings Errors		
X Channel pressure(s) exceed the range set for acceptable pressures. Please select another protocol or modify the chosen protocol on the next page.	The user customized protocol contains pressure values that exceed the max allowable pressure values designated to the device type being used.	The user must either adjust the PSI values for each channel on the Edit Run Settings Screen, adjust the protocol on the Edit Protocols – Edit Screen, or select a different protocol by returning to the Protocol Selection Screen.
Please enable at least one channel in order to start run.	No channels were enabled when "Start" was pressed.	The user must enable at least one channel on the Edit Run Settings Screen or return to Operator Selection Screen.
General Instrument Errors		
Input pressure too low	The input pressure is below the set target pressure. The compressed air source may not	The user should ensure the compressed air source is on, regulated to between 100 – 120

Error Message	Possible cause	Action
	be on, or is not delivering the correct amount of pressurized air to Infinity MTx.	PSI, the tubing is securely fastened between compressor and the Infinity MTx, there are no leaks in the external pneumatic air lines, and (for a cylinder of compressed air) contains enough capacity. If the problem persists, contact customer support.
Piston pressure too low	The piston pressure is below the set target pressure. The device will not properly seal during operation.	Remove the device from the Infinity MTx, remove the MPA and re-place the MPA in the sample loading tray. Reattempt the transfection protocol. If the problem persists, contact customer support.
Manifold cannot be lowered into position	The manifold is unable to be lowered properly. The device will not properly seal during operation.	Remove the device from the Infinity MTx, remove the MPA and re-place the MPA in the sample loading tray. Re-attempt the transfection protocol. If the problem persists, contact customer support.
Air tank pressure too low	The air tank pressure is below the set target pressure.	User should contact customer support.
Manifold not raised, unable to eject consumable	The manifold is not fully raised preventing the device from being ejected properly.	User should manually open the sample loading tray by pulling at the bottom of the tray. If the problem persists, contact customer support.
Run Message Errors		
Detected flow rate too high, liquid sample not present	Baseline air flow is too high, indicating the absence of sample in the indicated sample chamber.	After the remainder of the run is complete, ensure the sample is loaded in the device by checking the contents of the sample chambers in a biosafety cabinet. If empty, load sample chambers before attempting to run the transfection process.

Error Message	Possible cause	Action
Detected flow rate too low, leak detected	Baseline air flow is too low, indicating a liquid or air leak in the system/MPA in the indicated sample chamber.	Contact customer support to replace faulty device.
Clogged channel	Clogging was detected in the indicated sample chamber.	Before the next transfection process, ensure the sample cell density and/or volume is lowered and ensure proper controls are taken to not introduce large particulate contamination.
Channel time-out	The indicated sample chamber did not finish processing within maximum allowable amount of time.	Before the next transfection process ensure cell density and/or volume is lowered and ensure proper controls are taken to not introduce large particulate contamination.

Chapter 5: The Transfection Process

Principles of Operation

The CellFE Infinity MTx system is built upon the principles of microfluidic design. Using micron-scale structures, a newly discovered cell behavior of volume exchange for convective transfection (VECT) can be observed. This VECT phenomenon utilizes active transport generated in response to large magnitude deformations at ultrafast timescales in a manner that does not damage the cells[1,2]. On a biological level, transient cell membrane pores and active transport of payload into the cells are created by rapidly flowing

4A)

Image showing an overhead view of the microchannel. Similar to the overhead view of device as seen in the video on the CellFE website.

4B)

Image showing a side view of the microchannel. Similar to the side view (with the cells), seen in the video on the CellFE website.

mammalian cells through microfluidic channels with a series of abrupt, rectangular cross-section ridges[2] (Figure 4A, 4B).

The CellFE Infinity MTx works in tandem with the Infinity MPA to allow for the transfer of cell sample through the microfluidic channel. The CellFE Zephyr instrument supplies highly pressurized air to perform no-contact liquid transfer to each of the 8 sample wells in the Infinity MPA. This allows for the rapid transfection of a wide range of cell densities and total cell numbers. VECT has already proven useful in delivering large biological payloads, including mRNA[3], CRIPSR/Cas9 RNP, and Plasmid. Additionally, the system has shown efficacy in transfecting large inorganic nanoparticles (i.e., diameter of around 40 nm) into cells for in vivo imaging[4].

Performance Characteristics & Specifications

mRNA Delivery

CRISPR/Cas9 Delivery

Plasmid Delivery

Delivery of Other Biological Payloads

Trademarks patents and licenses

Various features of this product and corresponding processes may be protected by United States patents and described in various patent publications listed below. This listing is intended to satisfy the virtual patent marking provisions of U.S. patent law. This listing may not be all inclusive as new patents are expected to be granted and new patent applications are expected to be filed. Furthermore, patents and patent publications may include corresponding foreign and continuation cases. By listing patents in this document, CellFE makes no admission that other patents do not apply.

- US granted patent no. 8356714
- US granted patent no. 10,717,084
- US patent publication no. 2020330990
- CA patent publication no. 3082037
- CA granted patent no. 110198785
- EP patent publication no. 3538269
- JP granted patent no. 7084393
- US granted patent no. 11,198,127
- US patent publication no. 20220062904
- EP patent publication no. 3538270
- US granted publication no. 11268892
- US granted publication no. 11668638
- US patent publication no. 2021292700
- US patent publication no. 20220056393
- KR patent publication no. 20210148075
- US patent publication no. 20220298461
- US patent publication no. 20220213422
- PTO WO patent publication no. 2022/109113
- US registered trademark no. 6591175

Methods and Operation

VECT Parameters

The Infinity MTx is designed to operate within specific parameters. The values and limits for each parameter is listed below:

- Device design (range: 5.2 – 11)

- Operating pressure supply (5 – 80 PSI)
- Cell type
- Payload type
- Buffer type (recommended: basal cell culture media used for cell type of interest)

Operation Recommendations

- Wear gloves, laboratory coat, and safety glasses during the transfection.
- **Use high quality payloads (of DNA, mRNA, RNP) to ensure good transfection efficiency** (recommendations are provided below)**.**
- Follow the guidelines for cell preparation located on pages [pages]
- Use a fluorescent protein in expressing payloads as controls for optimization of the transfection protocol.
- Do not use each chamber within the Infinity MPA and/or optimizer kits more than once.
- Utilize only sterile materials and reagents.

Recommended Buffers

No proprietary cell buffers are required for the transfection process. However, transfection efficiency and viability of the transfection process is generally improved by utilizing basal cell culture medium (i.e., cell culture medium with no exogenous additives) during the transfection process.

DNA Quality and Amount

The effectiveness of the transfection process is highly dependent on the quality and concentration of the DNA employed in the process. It is advocated that the utilization of high-quality plasmid purified in a highly controlled manner is employed. Additional considerations are as follow:

- DNA should be reconstituted in deionized water or TE buffer at a concentration ranging from 1 to 5 ug/uL.
- It is advisable to keep the volume of added DNA solution below 10% of the total reaction volume.
- You may check the purity of the purified DNA preparation by measurement of the $A_{260/280}$ ratio. The recommended ratio should be at least 1.8.
- Plasmids up to approximately 20 kb should not be a problem. Using plasmids larger than 20 kb will most likely lower transfection efficiency.
- If frozen, thawing material slowly on ice for ~20 – 30 minutes is recommended.

NOTE: Do not precipitate DNA with ethanol to concentrate DNA. This may result in salt contamination which can lead to poor transfection efficiency and cell viability.

mRNA Quality and Amount

The effectiveness of the transfection process is highly dependent on the quality and concentration of the mRNA employed in the process. It is strongly recommended to utilize high-quality mRNA modified with a 5' capping solution to increase editing efficiency and reduce degradation by nucleases. Additional considerations are as follow:

- Starting mRNA concentration is 100 – 250 uM in nuclease-free water.
- Reduction of freeze-thaw cycles of the mRNA, significant amounts of freeze-thaw cycles can impact editing efficiency

- It is recommended to utilize nuclease-free environment conditions during the transfection process.
- It is advisable to keep the volume of added mRNA solution below 10% of the total reaction volume.
- If frozen, thawing material slowly on ice for ~20 – 30 minutes is recommended.

Controls

Determining Required Cell Amounts

The Infinity MTx can operate at a wide range of cell densities (1 – 5 x 10^7 cells/mL), but must be operated within a volume range of 30 – 100 µL for each cell sample chamber. Given this operational range, the user can determine the total number of cells to be processed through each channel (see table below). **NOTE:** The max throughput of a single chamber is 2.50 x 10^6 cells, if working with higher cell densities, volumes may need to be reduced and visa-versa. Post-transfection culture conditions can be based on the relevant parameters used.

Recommended Volumes and Cell Density for Processing

		Cells Density (Cells / mL)				
		1.00 x 10^7	*1.50 x 10^7*	*2.00 x 10^7*	*2.50 x 10^7*	*5.00 x 10^7*
Volume Processed (µL) per channel	*30*	3.00E+05	4.50E+05	6.00E+05	7.50E+05	1.50E+06
	40	4.00E+05	6.00E+05	8.00E+05	1.00E+06	2.00E+06
	50	5.00E+05	7.50E+05	1.00E+06	1.25E+06	2.50E+06
	75	7.50E+05	1.13E+06	1.50E+06	1.88E+06	
	100	1.00E+06	1.50E+06	2.00E+06	2.50E+06	

Preparation of Adherent Cells

1. Culture the necessary quantity of cells (70 – 90% confluency on the day of the transfection) in fresh, complete cell culture media in a flask 1 – 2 days before the transfection process. For the most optimal transfection procedure:
    a. Seed X cells [min] to Y cells [max] for each reaction chamber to be used on the Infinity MPA during the day of the transfection.
2. Warm up sample of complete cell culture medium (with serum to neutralize Trypsin), PBS (-Ca^{2+} / -Mg^{2+}), Trypsin/EDTA solution, and an aliquot of basal cell culture medium.
    a. For the quantity of basal cell culture medium use the following formula to determine the minimum amount necessary:

$$(\#\ of\ reactions\ to\ be\ run)x(volume\ per\ reaction\ [mL]) = (total\ volume\ of\ basal\ cell\ culture\ media\ [mL])$$

3. Aspirate the media from the cells, being careful not to aspirate the cells from the flask. Rinse the cells with the pre-warmed DPBS.
4. Add Trypsin/EDTA or TrypLE Express to the cells ensuring the entire base of flask is coated. Let cells sit in the flask for 1 – 2 minutes. Then neutralize the cells with required volume (ml) of complete cell culture medium containing serum.

5. Mix cells thoroughly using a large bore (10 mL serological pipette) and remove an aliquot for cell counting to identify the cell density and total cell amount.
6. Transfer the cells to a microfuge tube or conical, depending on the quantity required.
7. Centrifuge the cells at 100 – 400 x g for 5 – 10 minutes at room temperature.
    a. Centrifugation speed and time are dependent on the cell type.
8. Wash the cells with PBS (-Ca^{2+} / -Mg^{2+}), at centrifuge again with the same conditions used during the initial harvesting step.
9. Aspirate the PBS and resuspend the cells to a final cell density of 1 – 5 x 10^7 cells/mL using the basal cell culture medium. Use a pipette to mix the cell solution to achieve a single-cell suspension.
    a. Cell density can be adjusted depending on the desire of the operator.

NOTE: Avoid keeping the cells in suspension for more than 15 – 30 minutes at room temperature. Due to the nature of the container and the non-optimal temperature fluctuation, transfection efficiency and cell viability may be reduced.

10. Prepare post-transfection cell culture plates by filling wells of a 24-well plate with 0.5 mL of complete cell culture media. Pre-incubate plates in an incubator set to 37°C with 5% CO_2.

Preparation of Suspension Cells

1. Culture the necessary quantity of cells (~0.5 – 2 x 10^6 cells/mL on the day of the transfection) in fresh, complete cell culture media in a flask 1 – 2 days before the transfection process. For the most optimal transfection procedure:
    a. Seed X cells [min] to Y cells [max] for each reaction chamber to be used on the Infinity MPA during the day of the transfection.
2. Warm up sample of complete cell culture medium (with serum), PBS (-Ca^{2+} / -Mg^{2+}), and an aliquot of basal cell culture medium.
    a. For the quantity of basal cell culture medium use the following formula to determine the minimum amount necessary:

$$(\# \ of \ reactions \ to \ be \ run)x(volume \ per \ reaction \ [mL]) = (total \ volume \ of \ basal \ cell \ culture \ media \ [mL])$$

3. Mix cells thoroughly using a large bore (10 mL serological pipette) and remove an aliquot for cell counting to identify the cell density and total cell amount.
4. Transfer the cells to a microfuge or conical tube, depending on the quantity required.
5. Centrifuge the cells at 100 – 400 x g for 5 – 10 minutes at room temperature.
    a. Centrifugation speed and time are dependent on the cell type.
6. Wash the cells with PBS (-Ca^{2+} / -Mg^{2+}), at centrifuge again with the same conditions used during the initial harvesting step.
7. Aspirate the PBS and resuspend the cells to a final cell density of 1 – 5 x 10^7 cells/mL using the basal cell culture medium. Use a pipette to mix the cell solution to achieve a single-cell suspension.
    a. Cell density can be adjusted depending on the desire of the operator.

NOTE: Avoid keeping the cells in suspension for more than 15 – 30 minutes at room temperature. Due to the nature of the container and the non-optimal temperature fluctuation, transfection efficiency and cell viability may be reduced.

8. Prepare post-transfection cell culture plates by filling wells of a 24-well plate with 0.5 mL of complete cell culture media. Pre-incubate plates in an incubator, i.e., set to 37°C with 5% CO_2.

Setting up the Infinity MTx

1. Start the Infinity MTx by turning on the Power Switch (located at the back panel of the instrument) to the "on" position. The Instrument will power up.
2. Turn on the compressed air source, either an air compressor, house air, or a compressed N_2 gas tank. Before proceeding, ensure the output pressure has reached 100 PSI and has not exceeded 120 PSI.

Loading Sample into the Infinity MPA

1. In a sanitized biosafety cabinet, open the packaging containing the Infinity MPA that will be used for the transfection process.
2. Keeping the MPA steady, open the MPA lid by pushing up on the MPA lid latch.
3. Transfer the appropriate amount of payload into a sterile microfuge tube.
4. Add the prepared cells that are resuspended in basal cell culture media to the tube containing the payload and gently mix the components with a pipette.
5. Using a pipette load the desired volume of the cell and payload solution into the desired number of sample chambers.
6. Once complete, while keeping the body of the MPA steady, close the MPA lid and ensure the MPA lid latch has engaged.
7. Move the MPA over to the Infinity MTx.

Schematic/photo showing the key on the sample loading tray (overhead view with no device) and another overhead view of the MPA loaded into the sample loading tray

Running Samples in the Infinity MTx

1. Select and sign in as an Operator or an Administrator.
2. On the Insert MPA Screen, use the Eject Button to eject the sample loading tray. Carefully place the MPA (filled with samples to be transfected) into the sample loading tray. Once complete, press the Withdraw Button on the screen to retract the sample loading tray.
3. Select the desired protocol on the Protocol Selection Screen.
4. Make an adjustment to the protocol if desired, including disabling or adjusting the pressure of certain chambers, write run notes, add protocol information, or scan the ID of the device.
5. Once setup is complete, press the "Start" Button to initiate the transfection process and wait until all sample chambers have been run.

Schematic/photo showing the inlet and outlet side of the MPA with the lid on and the lid off

6. After the run has been completed, press the Eject Button to eject the sample loading tray, remove the MPA, and press the withdraw button to return to the Insert MPA screen.
7. Return to the biosafety cabinet with the MPA to perform the post-processing steps.

Removing Sample and Cell Culture

1. In the biosafety cabinet, while holding the body of the MPA steady, either:
   a. Remove the MPA lid pre gently pressing up on the MPA lid latch, or
   b. Remove the entire collection chamber lid by squeezing both collection chamber latches at the same time with the thumb and the pointer finger; ensure the collection chamber is held steady during this process.
2. The transfected cell sample will be pooled at the bottom of each chamber that was ran. To remove the sample, use a pipette to move the cell sample from the collection chamber to the prepared 24-well culture plate or similar. Repeat the process with all the samples desired.
3. Gently rock the culture system to ensure even distribution of cells. Incubate the culture, i.e., at 37°C in an incubator with 5% CO_2.
4. Dispose of the MPA in the biohazardous waste after use.

Infinity MTx Shutdown

1. If the Infinity MTx does not to be used further, turn off the source of compressed air.
2. Turn off the Infinity MTx by toggling the Power Switch to the “Off” position. The switch is located on the back panel of the instrument.

Assay Samples

1. Samples can be assayed according to the type of payload transfected into the cells (i.e., mRNA and/or transfections typically ~1-2 days after the transfection event, CRISPR/Cas9 gene edits typically ~3-5 days after transfection event).
2. Determine transfection results with the relevant type of assay (i.e., fluorescence microscopy, flow cytometry, or functional assay).

Cleaning and Maintenance

⚠ **CAUTION:** Proper cleaning and maintenance of the Infinity MTx are crucial to maintain its optimal performance. **NOTE:** Only utilize the cleaning and maintenance methods explicitly outlined in this section. The user must ensure strict adherence to the following guidelines:

- **DO NOT** employ decontamination or cleaning agents that could potentially create hazards due to reactions with any parts of the equipment or the substances contained within it.

- **DO NOT** perform any decontamination while the Infinity MTx is turned on. Turn off the Infinity MTx and the source of compressed air and ensure the Infinity MTx is adequately decompressed prior to any cleaning or maintenance.

- The Infinity MTx must be appropriately decontaminated in the following scenarios: a) if hazardous materials are accidentally spilled onto or into the equipment, and/or b) before scheduling service, repair, maintenance, trade-in, disposal, or the termination of a loan for the instrument at your facility.

- Before employing any cleaning or decontamination techniques that deviate from the manufacturer's recommendations, users should verify with the manufacturer that the proposed method will not inflict damage upon the equipment.

By meticulously adhering to these guidelines, you uphold the integrity of your biotech instrument's performance while safeguarding against any potential hazards or unintended consequences.

Clean the surfaces of the Infinity MTx using a slightly damp cloth or disposable laboratory paper. Do not use organic solvents or harsh chemicals to clean the Infinity MTx. The instrument can be cleaned by using a 70% ethanol solution.

In the case of a non-hazardous liquid spill inside or on the Infinity MTx, wipe the spill with a dry laboratory paper. In the case of a spill that could pose a biological hazard, utilize proper personal protective equipment (laboratory coat, safety glasses, and gloves) and abide by your lab's standard operating procedure(s) to clean up biological hazards.

For any other repairs and service contact, CellFE Biotech Technical Support.

WARNING: Do not perform any repairs or service by yourself to avoid damage to the Infinity MTx or yourself.

Operational Precautions & Limitations

Limitations of Infinity MTx System

- The maximal pressure that can be set as operational pressure is 90 PSI. Some compressed air sources may exceed or fall below the specified maximal operational pressure.
    - If the input pressure falls below the necessary 100 PSI threshold and the desired operating running pressure is set to 80 – 90 PSI, issues with cellular transfection and liquid transfer in the instrument may occur.
- In some cases, if the operating cell density is too high or the cell number exceeds the values suggested 2M cells per chamber, the Infinity MPA may not function with low running pressures (5 – 20 PSI). In this case the sample will not move from the inlet to outlet side and an error may occur.
- The Infinity MPA cannot fit or operate correctly with any third-party or in-house made consumable device. Only CellFE-produced Infinity MPA and/or Optimizer kits should be used with the Infinity MTx.

Limitations of CellFE Infinity MPA

- Each chamber within the Infinity MPA is intended for a single use only. Chamber re-use can lead to running errors, and negatively affect outcome cell transfection efficiency and/or cellular viability.
- The liquid level of each chamber in the Infinity MPA should not fall below 25 µL and is not to exceed 100 µL of sample volume.
- Cell densities should remain in the range of [indicate cell density range for a single sample]. While densities outside this range have been tested, processing outside of this range may lead to errors

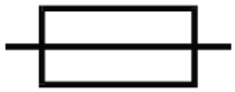
while running and/or lead to decreased outcome cellular transfection efficiency or cellular viability.

- Achievement of desirable results with all cell types is not guaranteed.
- It is recommended that thawed cells should be reconstituted and cultured for a minimum of one day. Results of transfecting cells immediately after thawing cells is not guaranteed.
- Achievement of desirable results with all payloads is not guaranteed.
- The Infinity MPA operates based on microfluidic design related to small channel geometries. Large, solid particles (i.e., metallic nanoparticles >1µm) may lead the MPA not working correctly.

Appendix A: Symbols Glossary

Symbol	Explanation
CE	The CE mark symbolizes that the product conforms to all applicable European Community provisions for which this marking is required. Operation of the Infinity Mtx Instrument is subject to the conditions described in this manual. The protection provided by the device may be impaired if the instrument is used in a manner not specified by the manufacturer.

	This product conforms to UL 61010-1, CAN/CSA C22.2 No.61010-1 “Safety Requirements for Electrical Equipment for Measurement, Control, and Laboratory Use, Part I: General Requirements.” Instruments bearing the ETL symbol are certified by ETLProduct Services to be in conformance with the applicable safety standard for the US and Canada.
	WEEE (Waste Electrical and Electronic Equipment) symbol indicates that this product should not be disposed of in unsorted municipal waste. Follow local municipal waste ordinances for proper disposal provisions to reduce the environmental impact of WEEE. This instrument meets European requirement WEEE Directive 2012/19/EU.
	Read operators manual before using the Infinity Mtx Instrument
	Potential biohazard.
	Warning. If ignored, can cause serious adverse reactions (death) and potential safety hazards (injury).
	Caution, If ignored, can trigger adverse reactions for safe and effective use of the device
I	Indicates the ON position of the instrument power switch.
O	Indicates the OFF position of the instrument power switch.

	Indicates the fuse rating for the device when replacing the fuse use one of the same type and rating as marked on the label.
∼	Indicates the AC power requirements of the device. Refer to the label to ensure that the facility can handle the requirements.
RUO	Research only use – "For Research Use Only. Not for use in diagnostic procedures."

Appendix B: Safety Information

For Research Use Only. Not for use in diagnostic procedures.

The Infinity MTx is designed to meet IEC 61010-1 Safety Standards. To ensure safe, reliable operation, always operate the Infinity MTx according to the instructions in this manual. Failure to comply with the instructions in this manual may create a potential safety hazard, and will void the manufacturer's warranty and void the safety standard certification. CellFE is not responsible for any injury or damage caused by use of this instrument when operated for purposes which it is not intended. All repairs and service should be performed by CellFE Inc.

Caution

- This Infinity MTx is not to be used for clinical investigation or clinical diagnostics.
- The Infinity MTx contains no user-serviceable parts. Do not attempt to open the enclosure. Contact CellFE Technical Support in case of problems with the instrument.
- The Infinity MTx contains a high voltage source. Do not attempt to open the enclosure.
- Do not operate the Infinity MTx in case of damage to the tray, enclosure, power cord, pressure line, interface screen, or device carrier.
- Biological specimens should be handled and disposed of with appropriate precautions as though biohazardous, including compliance to OSHA blood-borne pathogens standard (21 CFR 1910.1030).
- The Infinity MTx weighs 39.9 pounds (18.1 kg). Follow your institutional safety instructions for lifting heavy objects.
- Dispose of waste in accordance with federal, state, and local regulations. Prepare and use reagents and samples with appropriate good laboratory practice and caution. Wear appropriate personal protective equipment (e.g., lab coat, gloves, eye protection, disposable respirator).
- The Infinity MTx is connected to a high-pressure air source. Before unplugging the air line from the Infinity MTx, turn off the instrument and the source of compressed air, and ensure the source of compressed air has decompressed completely before proceeding.
- Turn off instrument when not in use.
- Avoid spilling liquids on the Infinity MTx. In case of spill, wipe clean with a lint-free dry wipe. Wipe may be wet with water or a dilute alcohol solution to clean residues.
- Ensure that the power supply input voltage matches the voltage available in your location
- Set the main switch on the power supply unit to OFF before connecting the power cord to the wall outlet.
- To avoid potential shock hazard, make sure that the power cord is properly grounded
- Set the main switch to OFF, unplug the power cord, and secure the pipette station before moving the device.

Safety

EU Directive 2014/35/EU – Low Voltage Directive

IEC/EN/UL/BS 61010-1 – Safety requirements for electrical equipment for measurement, control, and laboratory use – Part 1: General requirement.

CAN/CSA C22.2 No. 61010-1 – Safety requirements for electrical equipment for measurement, control, and laboratory use – Part 1: General requirement.

IEC/EN/UL/BS 61010-2-081 - Safety requirements for electrical equipment for measurement, control, and laboratory use – Part 2-081: Particular requirements for automatic and semi-automatic laboratory equipment for analysis and other purposes

CAN/CSA C22.2 No. 61010-1 - Safety requirements for electrical equipment for measurement, control, and laboratory use – Part 2-081: Particular requirements for automatic and semi-automatic laboratory equipment for analysis and other purposes

EU directives:

- Electromagnetic Compatibility Directive (EMC) 2014/30/EU
- Low Voltage Directive (LVD) 2014/35/EU
- Waste Electrical and Electronic Equipment (WEEE) Directive 2012/19/EU
- RoHS Directive 2015/863/EU (RoHS3)
- REACH Directive Reg. (EC) No 1272/2008
- Radio Equipment Directive (2014/53/EU).

Electromagnetic Compatibility (EMC)

Directive 2014/30/EU – European Union EMC Directive
IEC/EN/BS 61326-1 – Electrical Equipment for Measurement, Control and Laboratory Use – EMC requirements – Part 1: General Requirements
ICES – 003 – Information Technology Equipment (Including Digital Apparatus)
ETSI EN 301 489-1 V2.2.3 (2019-11), ElectroMagnetic Compatibility (EMC)
standard for radio equipment and services;
Part 1: Common technical requirements; Harmonised Standard for ElectroMagnetic Compatibility

ETSI EN 301 489-17 V3.2.4 (2020-09) ElectroMagnetic Compatibility (EMC)
standard for radio equipment and services;
Part 17: Specific conditions for
Broadband Data Transmission Systems; Harmonised Standard for ElectroMagnetic Compatibility

USA EMC compliance statement

Class A Digital Device

This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications.

Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at their own expense.

Canada EMC compliance statement

CAN ICES-003(A) / NMB-003(A)

This Class A device complies with Canadian ICES-003
Cet Clase A appareil est conforme à la norme NMB-003 du Canada.

RFID Reader

This device contains a DLP-RFID2D module

Frequency : 13.56 MHz
ERIP (Power) : 0.0329 mW

RFID reader FCC statement

Operation is subject to the following two conditions: (1) This device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesirable operation.

Contains Transmitter modules FCC ID: SX9RFID2

Industry Canada

Contains transmitter module IC: 5675A-RFID2

Contient le module émetteur IC : 5675A-RFID2

This device complies with Industry Canada license exempt RSS standard(s). Operation is subject to the following two conditions:

(1) this device may not cause interference, and

(2) this device must accept any interference, including interference that may cause undesired operation of the device.

Cet appareil est conforme aux CNR d' Industrie Canada applicables aux appareils radio exempts de licence. L' exploitation est autorisée aux deux conditions suivantes:

(1) l' appareil ne doit pas produire de brouillage, et

(2) l' utilisateur de l' appareil doit accepter tout brouillage radioélectrique subi, même si le brouillage est susceptible d' en compromettre le fonctionnement.

Environmental Requirements

Condition	Acceptable range
Operating Environment	Temperature: 15 to 30°C (59 to 86°F) Humidity: 20 to 80% RH (noncondensing)
Storage and transport conditions	Temperature: -30 to 60°C (-22 to 140°F) Humidity: 20 to 80% RH (noncondensing)

Electromagnetic interference	Do not use this device in close proximity to sources of strong electromagnetic (EMC) radiation (for example, unshielded intentional RF resources), as EMC radiation might interfere with the proper operation of the device.

Biological hazard


WARNING! Potential Biohazard – The surface maybe considered biohazard. Use appropriate decontamination methods when working with biohazards. The following references provide general guidelines when handling biological samples in laboratory environment.

· U.S. Department of Health and Human Services, Biosafety in Microbiological and Biomedical Laboratories (BMBL), 6th Edition, HHS Publication No. (CDC) 300859, Revised June 2020
www.cdc.gov/labs/pdf/CDC-BiosafetymicrobiologicalBiomedicalLaboratories-2020-P.pdf

· Laboratory biosafety manual, fourth edition. Geneva: World Health Organization; 2020 (Laboratory biosafety manual, fourth edition and associated monographs)
www.who.int/publications/i/item/9789240011311

This product complies with the safety requirements outlined in IEC 61010-2-081 for laboratory equipment. It may come into contact with biohazardous materials during its use. To ensure safe biohazardous waste disposal and overall product safety, please adhere to the following guidelines:

1. **Biohazard Identification:** Always clearly label and identify any materials or specimens that may contain biohazardous agents in accordance with your organization's biohazard safety protocols and applicable regulations.
2. **Material Safety Data Sheets (SDS or MSDS):** Access and familiarize yourself with Material Safety Data Sheets (SDS or MSDS) for any chemical substances or biohazardous agents used in conjunction with this product. MSDS provides critical information on the properties and safety precautions associated with these materials. Ensure that all users are aware of the location of MSDS and how to interpret them.
3. **Personal Protective Equipment (PPE):** When handling biohazardous materials or substances identified in the MSDS, wear the appropriate personal protective equipment (PPE) as recommended by the MSDS and your institution's safety guidelines. This includes gloves, lab coats, safety goggles, masks, or other protective gear specified.
4. **Biohazard Waste Containers:** Use appropriate biohazard waste containers designed for the safe collection and disposal of biohazardous materials. These containers should be clearly labeled with biohazard symbols and warnings.
5. **Decontamination:** Ensure that all surfaces and equipment that come into contact with biohazardous materials are thoroughly decontaminated following your institution's established decontamination procedures.
6. **Disposal:** Dispose of biohazardous waste in strict accordance with local, national, and international regulations governing biohazardous material disposal, as specified in the MSDS and your institution's waste disposal protocols.
7. **Emergency Response:** Familiarize yourself with emergency response procedures pertaining to biohazardous materials, as outlined in the MSDS and your institution's safety guidelines.
8. **Training:** It is essential that all individuals using this equipment receive proper training in biohazard safety practices, disposal procedures, and the utilization of MSDS.

9. **Support:** For inquiries related to biohazard safety, disposal, and MSDS specific to this product, please contact the manufacturer at the address provided above For information regarding safe disposal of materials please contact your local regulatory agency,

WEEE Directive Statement

Electronic and Electrical Equipment placed onto the market in Europe are marked with the crossed-out wheeled bin symbol to indicate that they are covered by the WEEE Directive (2012/19/EU), which imposes a number of obligations on producers of EEE including obligations relating to the financing of the take-back treatment and recycling of end of life equipment (WEEE).

To ensure proper disposal and recycling of this product, please follow the guidelines provided below:

1. End-of-Life Disposal: Do not dispose of this product with regular household waste. It is your responsibility to properly dispose of this product at the end of its life cycle.
2. Collection and Recycling: Contact your local authorities or recycling centers to find out about the designated collection points for electronic waste. In some regions, special collection facilities may be available for WEEE products.
3. Separation of Batteries: If this product is equipped with batteries, remove them before disposal and recycle them separately in accordance with local regulations.
4. Environmental Protection: By disposing of this product correctly, you are contributing to environmental protection and conserving natural resources.
5. Penalties may be applicable for incorrect disposal of this waste, in accordance with your national legislation.

For additional information on WEEE compliance and recycling options, please contact your local regulatory agency or the manufacturer at the address provided in this manual .

Proposition 65 warning

This product can expose you to chemicals including Nickel (metallic) which are known to the State of California to cause cancer or birth defects or other reproductive harm. For more information, visit www.P65Warnings.ca.gov.

Appendix C: Troubleshooting

Problem	Cause	Solution
No Power (the display is off when the Power Switch is in the "On" position)	Power cord is not properly plugged into the instrument	Check the connection of the power cord to the instrument or for any damage on the power cord. **NOTE:** Make sure to disconnect Power cord prior to its inspection.
Infinity MTx cannot detect the MPA	The device is incorrectly loaded into the sample loading tray or there is no device present.	Remove MPA from the sample loading tray, re-insert ensuring protrusion of device meets groove present on the sample loading tray.
	The Infinity MTx's presence sensor is unable to determine the presence of the MPA.	Clean the presence sensor on the sample loading tray with 70% ethanol solution and wipe off to ensure sensor is dry **NOTE:** Presence sensor is small window on the inside of the sample loading tray.
Error messages	See "Infinity Status and Run Error" section [pages]	See "Infinity Status and Run Error" section [pages]
Sample or samples do not process (protocol is unable to run)	No Pressure connected to the instrument	Plug in pressure line to instrument, make sure pressure line is correctly seated in compressed air inlet on the back of the instrument
	Pressure is present but the minimum required pressure cannot be achieved	Check regulator upstream of the Infinity MTx to ensure that the pressure is set between 100 and 120 PSI. Double check compressed air source to ensure a minimum of 100 PSI can be achieved.
	Air leak in line connecting compressed air source and Infinity MTx	Check line connecting compressed air source and the Infinity MTx. Observe for hissing noise while Infinity MTx is under pressure. If leak is found replace tubing with new tubing.
	Air leak within instrument prevents insufficient pressure supply	Pneumatics system in Infinity MTx may be damaged or dislodged. **NOTE:** This operation is to be performed by CellFE technical support team only.

Problem	Cause	Solution
Sample or samples do not process (protocol runs but no sample is processed)	MPA is not properly being sealed during transfection process	Transfer samples to a new, unused MPA. Discard old MPA. Contact CellFE Biotech [help email] for replacement MPA
	Device is contaminated with large particles	Ensure quality of mediums and environment when preforming the transfection process. Large particulates present in the solution may immediately clog the MPA.
Liquid leak after processing sample	Volume processed is too high	Reduce the amount of volume of each chamber to less than 100 uL.
	Defective consumable device	If the leak is spreading to the bottom of the MPA, determined by the presence of liquid in sample loading tray, transfer unprocessed samples to a new, unused MPA, discard old MPA. Contact CellFE Biotech [help email] for replacement MPA
Low recovery volume after transfection process	Volume processed is too low	Increase the amount of volume of each to more than 25 uL.
	Leak is occurring or sample is not processing fully.	See “liquid leak after processing sample” or “Sample does not process fully, or samples are taking a very long time to process” solutions in troubleshooting section.
	High operational pressure is aerosolizing media with certain applications	Reduce operational pressure on the Infinity MTx software during a run. While the system is able to process high pressure doing so with certain applications may displace the cell sample.
Sample does not process fully, or samples are taking very long time to process	Sample cell density or sample volume is too high	Reduce the cell density in the sample or the volume that is being used to transfect. Too many cells may clog the MPA.
	Undesired large particulates in media	Ensure quality of medias involved in cellular transfection process. Thoroughly remove any particles from cells before transfecting
	Multiple uses of MPA sample chambers	Do not use each sample chamber more than once, Process samples using a new unused sample chamber.

Problem	Cause	Solution
	Solution is too viscous to process due to cell resuspension media or payload	Utilize basal cell culture media for cell resuspension prior to the transfection. Mitigate high viscosity of certain payloads by reducing stock concentration, salt content, or volume of payload used for the transfection.
Consistent damage to the MPAs during operation (i.e., cracks or fractures on MPA)	Issues with sample loading tray or internal compression piston	Contact CellFE Biotech [help email] for service to the Infinity MTx and replacement MPA's
Low transfection efficiency	Poor optimization of transfection parameters	Perform testing and optimization of device type and PSI for specific cells. Utilize optimizer kits to identify optimal MPA and test range of PSIs (i.e., 20, 40, 60, 80, etc.) for application. See Optimization protocol section.
	Cell density too high or low	Optimize processing cell density for specific type of application. Recommended range for the Infinity MTx is 1 – 5 x 10^7 cells/mL
	Payload concentration too low	Optimize the payload concentration for the specific application. May need to alter depending on desired cell density that is used or the size of the payload that is desired to be delivered.
	Poor payload quality	Use high quality, DNA, mRNA, or RNP/sgRNA for transfection.
	Mycoplasma contaminated cells	Test cell cultures for *mycoplasma* contamination. If present, dispose of current culture and start new cell culture.
	Nuclease contamination	Sterilize working space with nuclease destroying reagents. Use nuclease-free pipette tips and tubes for sample preparation.
Low cell survival rate	Specific MPA is out of range for cell type being transfected	Use different MPA design intended for use with specific cell type. Cell viability after transfection may diminish if MPA design is meant for smaller cells. **NOTE:** Perform testing and optimization of device type. See Optimization protocol section.
	Operational pressures out of range for cell type being transfected.	Reduce operational pressure being used.

Problem	Cause	Solution
		NOTE: Perform testing and optimization of operational pressure. See Optimization protocol section.
	Poor payload quality	Use high quality, DNA, mRNA, or RNP/sgRNA for transfection.
	Payload concentration is too high	Optimize the payload concentration for the specific application. May need to alter depending on desired cell density that is used or the size of the payload that is desired to be delivered.
	Cells are damaged before transfection	Observe culture before transfection. Ensure good confluency and viability and that passage number is low. Treat cells gently before transfection event, centrifuging at low speeds and mixing cells gently with a pipette.
	Multiple uses of MPA sample chambers	Do not use each sample chamber more than once, Process samples using a new unused sample chamber.
Irreproducible transfection efficiency	Inconsistent cell confluency, passage number, or culturing methods	Use cells at low passage number (describe low). Transfect cells at similar confluences. Utilize same techniques and reagents for cell cultures (i.e., culture medias, activation reagents, serums, etc.)
	Multiple uses of MPA sample chambers	Do not use each sample chamber more than once, Process samples using a new unused sample chamber.

Appendix D: Optimization Procedure

The process of achieving optimal transfection efficiency and cell viability, known as VECT, hinges upon the careful consideration of two key parameters: the microfluidic channel geometry and the input pressure applied to the system. These parameters together determine the flow velocity of cells within the microchannel. In certain cases, the initial outcomes of the VECT process may call for the fine-tuning of these parameters, particularly in relation to the specific cell type and payload in use.

Effective optimization of the VECT parameters empowers the Infinity MTx to perform consistent genetic manipulation across a spectrum of cell types and payloads. This is underpinned by the understanding that much of the optimization groundwork is laid during the microfluidic design phase.

Complementing the standard MPA, which offers 8 sample chambers with a single microfluidic design, the Infinity MTx introduces Optimization kits. These kits encompass 4 distinct microfluidic designs, each represented in duplicate across the 8 sample chambers of the device. The Optimization kits provide an invaluable platform for assessing the most suitable MPA for the specific cell type and payload at hand, with the goal of achieving optimal transfection efficiency and cell survival. The procedural steps leading to the determination of the ideal MPA variant and operating pressure are outlined below:

1. **Preliminary setup:** Execute the procedures detailed in the "Operating the Infinity MTx" section, spanning from cell preparation to the configuration of the Infinity MTx apparatus.
    a. **NOTE**: It's crucial to ensure an adequate volume of cell samples and payload is prepared, as all sample chambers within the Optimization kit will be utilized.
2. **Uniform loading:** Load the 8 sample chambers of the Optimization kit with an identical mixture of cells and payload.
3. **Device loading:** Integrate the device into the Infinity MTx system. During the protocol selection phase, opt for the "Initial Optimization Protocol."
4. **Execution and analysis:** Run the sample chambers within the Infinity MTx system as dictated by the chosen protocol.
5. **Post-processing**: Following completion of the experimental run, subject the cell samples to appropriate culturing conditions, catering to their specific requirements.
    a. **NOTE**: It's advised to label the cell samples during culturing, referencing their respective locations within the 8 sample chambers.
6. **Analysis and identification**: Perform a comprehensive analysis of the cell samples, evaluating both transfection efficiency and cell viability. With reference to the map insert accompanying the Optimization kit and the results of the cell analysis, pinpoint the sample chambers that yielded the most promising outcomes.
7. **Next steps:** Should the need arise, proceed to place an order for the pertinent and specific MPAs to facilitate further experimentation.
8. **Pressure optimization:** Once the optimal MPA configuration has been established, proceed to fine-tune the supply pressure. Repeat the optimization process using the chosen MPA, testing with no fewer than 4 distinct supply pressures.
    a. **NOTE:** While operators possess the flexibility to dictate specific supply pressures, it is recommended to initiate testing with pressures during the initial pressure optimization phase. Subsequent increments can be finetuned as required.

By adhering to this comprehensive optimization procedure, operators can confidently determine the most effective microfluidic design and supply pressure for their specific cellular manipulation needs, ultimately fostering enhanced transfection efficiency and cell survivability.

Appendix E: Technical Data

Input Voltage:	110 – 240 VAC, Frequency 50/60 Hz
Current	5.9-2.7A
Protective Earthing:	Class I
Input Pressure:	90 to 120 PSI
Maximum Input Pressure:	120 PSI
Air line Maximum Output Pressure:	90 PSI
Air line Minimum Output Pressure:	5 PSI
Operating Altitude:	Up to 2,000 meters above sea level
Operating Temperature:	15°C to 30°C (59 to 86°F)
Maximum Relative Humidity:	80% for temperatures up to 31°C decreasing linearly to 50% relative humidity at 40°C
Degree of ingress Protection:	IPx0
Instrument Type:	Benchtop Unit, for indoor use only
Max Device Dimensions:	11.75" X 11.75" X 10.63"
Device Weight:	39.9 pounds (18.1 kg)

The Infinity MTx has not been verified using all standard nonhazardous laboratory reagents. **WARNING:** Do not use instrument and device with hazardous or potentially explosive reagents or materials such as organic solvents or combustible liquids.

Appendix F: Documentation and Support

Customer and Technical Support

Limited Product Warranty

References

1. Liu, A. *et al.* Microfluidic generation of transient cell volume exchange for convectively driven intracellular delivery of large macromolecules. *Mater. Today* **21**, 703–712 (2018).

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3. Loo, J. *et al.* Microfluidic transfection of mRNA into human primary lymphocytes and hematopoietic stem and progenitor cells using ultra-fast physical deformations. *Scientific Reports* **11**, 21407 (2021).

4. Nejadnik, H. *et al.* Instant labeling of therapeutic cells for multimodality imaging. *Theranostics* **10**, 6024–6034 (2020).